ICC-1 Vertical subjects: high yield

Year 3, 2026-2027. One page per lecture, then one summary per Specification Grid topic.

Part 1: lectures

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5. Pathology of COPD, asthma and ILD [004] and lung pathology [033]
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30. Clinical therapeutics: cardiology [VM 110]
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Part 2: topic summaries

1. Pathology
2. Radiology
3. Mental health
4. Ophthalmology
5. Dermatology
6. Poisoning and toxicology
7. Use of medicine
8. Biochemistry

PART 1: LECTURE

Disorders of blood flow [VM 035] (parts 1 and 2)

Topic: Pathology

Hyperemia, congestion, edema

- Hyperemia = active arteriolar dilatation (exercise, inflammation). Congestion = passive venous engorgement from impaired outflow, with edema.
- CVC lung (mitral stenosis, left HF): heavy, wet, frothy lungs; late = heart failure cells (hemosiderin macrophages) + fibrotic septa.
- CVC liver (right HF): nutmeg liver = congested central veins + centrilobular necrosis (red) and peripheral steatosis (yellow); late fibrosis.
- CVC spleen (main cause: cirrhosis): big heavy spleen, Gamna-Gandy fibrosiderotic nodules.
- Edema mechanisms: raised hydrostatic pressure (HF, venous obstruction), low plasma oncotic pressure (nephrotic, cirrhosis, malnutrition), increased permeability (inflammation = exudate), Na retention, lymphatic obstruction.
- Exudate: protein > 3 g%, SG > 1015, clots, cellular. Transudate: the opposite.
- Lymphedema: Milroy, filariasis, peau d'orange (breast cancer), post-mastectomy; non-pitting. Renal edema starts periorbital.

Thrombosis

- Virchow's triad: endothelial injury, altered flow (stasis, turbulence), hypercoagulability.
- Stasis = most common cause of venous thrombosis (DVT after immobilization, right HF). Turbulence: aneurysms, AF.
- Factor V Leiden: Va resistant to protein C; young recurrent venous thrombosis. Acquired: OCPs, pregnancy, surgery, Trousseau (pancreatic cancer).
- Lines of Zahn = layers of platelets/fibrin and RBCs; formed in flowing blood during life (heart, aorta).
- Venous thrombi almost always occlude and easily embolize; arterial thrombi form on ruptured plaques.
- Fate: lysis, embolization, propagation, organization and recanalization, incorporation into wall.

Embolism and infarction

- PE: massive = sudden death; medium = no effect in healthy lung but hemorrhagic infarct in congested lung (MS, left HF); repeated small = cor pulmonale.
- Paradoxical embolus: venous embolus crosses septal defect to systemic arteries.
- Systemic emboli (mural thrombus, aneurysm, plaque, vegetations) lodge in end arteries.
- Infarct = coagulative necrosis, except brain (liquefactive). Pale wedge-shaped: heart, spleen, kidney. Red: venous occlusion (torsion, strangulation), lung, dual supply, congested tissue.

Gangrene

- Dry: arterial occlusion, veins patent; mummified, slow putrefaction, mild toxaemia, clear demarcation and separation.
- Moist: artery + vein (crush, tourniquet, bed sores, diabetic, intestine); rapid putrefaction, severe toxaemia, no separation line.
- Gas gangrene: clostridia in deep wounds; crepitus.

Source: VM 035 Disorders of blood flow part 1 and part 2 (2026-2027)

PART 1: LECTURE

Atherosclerosis, aneurysms and ischaemic heart disease pathology [037a, 037b]

Topic: Pathology

Atherosclerosis: lesions and pathogenesis

- Patchy intimal thickening by lipid surrounded by connective tissue (atheroma); mainly large and medium muscular arteries.
- Fatty streak: earliest lesion; LDL + lipid-laden macrophages; appears in infancy; flat, asymptomatic, reversible.
- Simple plaque: fibrous cap (SMCs, collagen) over a soft core (cholesterol clefts, necrotic debris, foam cells); neovessels at the periphery; common at bifurcations.
- Sites in descending order: lower abdominal aorta, coronaries, popliteal, internal carotid, circle of Willis; upper limb usually spared.
- Complicated plaque: rupture/ulceration (thin cap) with thrombus and emboli; haemorrhage into plaque from neovessels; media atrophy causing aneurysm.
- Response to injury: endothelial dysfunction, LDL entry, monocytes become macrophages, oxidized LDL engulfed = foam cells, growth factors drive SMC proliferation.
- Major risk factors: age, male, family history, genetics, hyperlipidaemia, smoking, hypertension, diabetes.

Consequences

- Coronary: angina, MI. Femoral: intermittent claudication (partial), dry gangrene (complete). Cerebral: TIA, stroke. Mesenteric: abdominal claudication, bowel infarction.

Aneurysms

- True (all 3 layers), false (walled-off extravascular haematoma), dissection (blood in media).
- Atherosclerotic AAA: elderly, fusiform, below renal arteries, with mural thrombus. Elective surgery if > 5 cm or enlarging; emergency surgery for rupture/dissection (50% mortality).
- Berry: circle of Willis; media weakness; ADPKD, Ehlers-Danlos IV, Marfan, smoking, HTN; causes SAH.
- Charcot-Bouchard: lenticulostriate vessels; hypertension. Syphilitic: ascending aorta, endarteritis of vasa vasorum. Mycotic: infection, often septic emboli.
- Dissection: intimal tear, blood tracks in media; HTN (men 40-60), Marfan; not atherosclerosis. Type A ascending, Type B distal to arch.

Ischaemic heart disease and MI

- 90% due to coronary atherosclerosis; significant stenosis > 75%.
- Transmural (occlusion, commonest) vs subendocardial (inner third; lysed thrombus or shock, circumferential).
- Necrosis begins in subendocardium; contractility lost in 60 s, necrosis by 30 min.
- Timeline: 1-3 d coagulation necrosis + neutrophils; 3-7 d yellow centre, hyperaemic rim; 10-14 d granulation tissue; > 2 mo dense scar.
- Complications: arrhythmia, acute failure, rupture (2-4 d, tamponade), papillary muscle rupture (MR), pericarditis (day 2-3), mural thrombus, Dressler's, ventricular aneurysm (late).

Source: 037a ATH and aneurysms; VM 037b Pathology of IHD (2026-2027)

PART 1: LECTURE

Cardiomyopathy and myocarditis histopathology [VM 036]

Topic: Pathology

Definition

- Cardiomyopathy (CM) = intrinsic myocardial defect or myocardial disease from a non-cardiovascular cause.
- Damage from hypertension, coronary disease, valvular or congenital disease, or myocarditis is not strictly CM ("ischaemic CM" is a clinical label).

Dilated CM (most common)

- All chambers dilated; big globular heart, thin balloon-like walls; dilatation masks hypertrophy.
- Systolic dysfunction (hypokinetic heart).
- Microscopy: hypertrophied fibres with dark nuclei, interstitial fibrosis, T-lymphocytes.
- Causes: genetic 30-40% (dystrophin: Duchenne/Becker), alcohol, anthracyclines (adriamycin), previous myocarditis, nutritional, haemochromatosis, peripartum, sarcoidosis.
- Effects: progressive heart failure, mural thrombus and embolism; 50% die within 2 years; transplant.

Hypertrophic CM

- Massive hypertrophy without dilatation; asymmetric septal hypertrophy; banana-shaped cavity; hyperkinetic LV; diastolic dysfunction.
- Mitral leaflet hits septum in systole: outflow obstruction, subaortic fibrosis.
- Microscopy: myocyte hypertrophy, pleomorphic nuclei, myocyte disarray, interstitial fibrosis.
- Genetic (50% familial AD): beta-myosin heavy chain most often, then myosin-binding protein C, troponin T.
- Commonest cause of sudden death in young people during vigorous exercise; syncope, AF, mitral endocarditis. Treatment: medical, pacing, septal myomectomy, alcohol ablation.

Restrictive and other CM

- Restrictive: infiltrated firm myocardium; diastolic dysfunction; near-normal systolic function; normal-sized ventricles; biatrial dilatation.
- Causes: amyloidosis, radiation fibrosis, sarcoidosis, metastatic tumour, storage disorders.
- Amyloid: pale pink on H&E; Congo red orange-red with apple-green birefringence. AL = light chains (plasma cell neoplasm); AA = SAA from liver (chronic inflammation, TB); ATTR = familial/old age; AANF = senile atrial.
- Endomyocardial fibrosis: restrictive; children, tropics. ARVC: RV fibrofatty replacement, ventricular arrhythmias.

Myocarditis

- Myocardial inflammation, usually with fever; often systolic dysfunction; may have pericarditis.
- Causes: Coxsackie A/B (most common significant cause), other viruses, bacteria, protozoa; immune (rheumatic fever, SLE, drugs, transplant rejection); giant cell myocarditis.
- Viral: dense T-lymphocyte infiltrate with necrosis. Outcome: no symptoms to heart failure or sudden death.

Source: VM 036 Histopathology of cardiomyopathy and myocarditis (2026-2027)

PART 1: LECTURE

Pathology of hypertension [VM 090a]

Topic: Pathology

Definition and causes

- Hypertension = persistent resting blood pressure above 140/90.
- Primary (essential) = 95%, no obvious cause.
- Secondary: renal (renal artery stenosis, glomerulonephritis, polycystic kidney, chronic pyelonephritis); endocrine (primary aldosteronism, Cushing, steroids, phaeochromocytoma, thyroid, acromegaly, pre-eclampsia); coarctation; neurologic.

Benign vs malignant

- Benign: commonest; gradual moderate rise; no symptoms; slow organ damage.
- Malignant (5%): rapid rise, usually > 200/120, symptomatic organ damage; de novo (young Afro-Caribbean men) or on top of benign (accelerated).

Benign hypertension: pathology

- Vessels: accelerated atherosclerosis; wall degeneration leading to aortic dissection and cerebral haemorrhage.
- Hyaline arteriolosclerosis: acellular pink thickening with narrow lumen (plasma leakage + extra matrix).
- Fibroelastic hyperplasia: small arteries; thick intima/media with reduplicated internal elastic lamina.
- Benign nephrosclerosis: small kidneys, finely granular surface, adherent capsule, poor corticomedullary demarcation, more peripelvic fat.
- Microscopy: hyaline arterioles, atrophic hyalinized glomeruli, atrophic tubules plus dilated tubules of working glomeruli. Clinically proteinuria and chronic renal failure.
- Heart: concentric LVH, ischaemic heart disease, later dilatation and LVF.

Malignant hypertension: pathology

- Malignant arteriolosclerosis: fibrinoid necrosis + hyperplastic (onion-skin) arteriolosclerosis from PDGF.
- Malignant nephrosclerosis: normal size (shrunken if accelerated), smooth flea-bitten surface, capsule strips easily.
- Microscopy: glomerular necrosis and haemorrhage.
- Heart: only mild LVH (short course).

Complications

- Vessels: atherosclerosis, aortic dissection, aneurysm rupture. Brain: cerebral (basal ganglia) haemorrhage.
- Heart: LVH, left ventricular failure (pulmonary oedema, heart failure cells, hypoxic encephalopathy), sudden arrhythmic death.
- Left heart failure symptoms: dyspnoea, orthopnoea, paroxysmal nocturnal dyspnoea.

Source: VM 090a Pathology of hypertension (2026-2027)

PART 1: LECTURE

Pathology of COPD, asthma and ILD [004] and lung pathology [033]

Topic: Pathology

Emphysema and chronic bronchitis

- Emphysema: permanent enlargement of air spaces distal to terminal bronchioles, wall destruction without fibrosis.
- Centriacinar: commonest symptomatic type; respiratory bronchioles; upper lobes; smokers.
- Panacinar: whole acinus; lower lobes; alpha-1 antitrypsin deficiency.
- Paraseptal: subpleural distal acinus, upper lobes; spontaneous pneumothorax. Irregular: commonest overall, with scarring, asymptomatic.
- Pathogenesis: smoke attracts neutrophils/macrophages; elastases and oxidants beat anti-elastases (alpha-1 AT).
- Pink puffer (emphysema): barrel chest, dyspnoea, good oxygenation. Blue bloater (+ bronchitis): infections, CO2 retention, cyanosis. Death: respiratory or right heart failure.
- Chronic bronchitis: productive cough 3 months in 2 consecutive years. Gland hypertrophy = hypersecretion; small airway disease = obstruction.

Asthma and bronchiectasis

- Asthma: reversible obstruction, eosinophilic inflammation, smooth muscle hypertrophy, mucus.
- Atopic (commonest): type I IgE, Th2 (IL-4, IL-5); positive skin test. Non-atopic: infection, pollutants.
- Plugs: Curschmann spirals (shed epithelium), Charcot-Leyden crystals (eosinophils). Remodelling: sub-basement membrane fibrosis, goblet cells, muscle hypertrophy.
- Status asthmaticus: attack lasting days, hypercapnia, acidosis, may kill.
- Bronchiectasis: permanent bronchial dilatation (muscle/elastic loss) from infection or obstruction; bilateral lower lobes; copious purulent sputum; CF commonest inherited cause. Complications: cor pulmonale, brain abscess, amyloidosis.

Pneumoconiosis and ILD

- ILD: interstitial inflammation and fibrosis; restrictive pattern.
- 1-5 micrometre particles most harmful; macrophage is key cell.
- Coal: anthracosis, upper-lobe coal nodules, progressive massive fibrosis. Silicosis: upper-lobe whorled collagen nodules, weakly birefringent; TB risk.
- Asbestosis: lower-lobe subpleural fibrosis, asbestos bodies; pleural plaques (commonest); mesothelioma risk.
- Sarcoidosis (type IV): hilar nodes, raised ACE/Ca; compact non-caseating granulomas along lymphatics; asteroid and Schaumann bodies.
- Hypersensitivity pneumonitis (type III + IV): farmer's/bird fancier's lung; BAL lymphocytosis, low CD4/CD8; poorly formed peribronchiolar granulomas.
- IPF: men 60+; fibrosis, honeycomb lung; dry cough, clubbing, crackles.

Pneumonia and immune injury

- Lobar (pneumococcus): congestion, red hepatization (3-4 d), grey hepatization (1 wk), resolution.
- Bronchopneumonia: patchy bilateral basal, extremes of age. Interstitial (viral, mycoplasma): septal lymphocytes, empty alveoli.
- Goodpasture (type II): anti-basement membrane IgG, linear; haemoptysis + haematuria.

Source: 004 COPD, asthma and ILD; C&B 033 Lung pathology (2026-2027)

PART 1: LECTURE

Pathology of pulmonary tuberculosis [010]

Topic: Pathology

Organism and staining

- Human type → pulmonary TB (droplet, most common route); bovine type (raw milk) → tonsil and intestinal TB in children; contact → skin.
- Lipid-rich wall: mycolic acids (60-90 C), cord factor (cytotoxin), sulfolipids (block phagolysosome fusion). Resists stains, many antibiotics, acids/alkalis; survives in macrophages.
- Ziehl-Neelsen: hot carbol-fuchsin → acid-alcohol → methylene blue. Slender, curved, red acid-fast bacilli.
- Risk: overcrowding, poverty, low resistance (DM), butchers, health care workers.

Granuloma (tubercle)

- Bacillus survives inside macrophage → presented to Th1 cells → IFN-γ (+TNF-α) activates and recruits macrophages; macrophage IL-12 boosts Th1.
- Tubercle: central caseous necrosis (homogeneous pink, cheese-like grossly), epithelioid cells, Langhans giant cells (horseshoe nuclei), peripheral lymphocytes and fibrosis.

Primary TB

- Unsensitised host. Ghon complex = Ghon focus (subpleural, middle third of lung, 1-1.5 cm, caseating) + lymphangitis + caseous hilar lymphadenitis.
- Fate: healing by fibrosis (commonest); direct spread to lung/pleura; blood spread (few bacilli destroyed, moderate → single organ, many → disseminated); later reactivation.

Secondary TB

- Sensitised host: reactivation (usual) or re-infection. Apices of upper lobes (high O2 tension); hypersensitivity → walling off → cavitation. Initial focus < 2 cm.
- By resistance: good → healed fibrotic apex; some → chronic fibro-caseous; very low → acute TB (whole-lobe caseation, no fibrosis); blood spread → miliary.
- Progressive TB (elderly, immunosuppressed): cavitating TB (ragged cavity; vessel erosion → haemoptysis; bronchial drainage = open TB), TB bronchopneumonia (airway spread, poor prognosis), miliary (< 2 mm foci, "galloping consumption"), single organ TB.
- Systemic miliary: liver, spleen, marrow, adrenals, meninges, kidney, genital tract; poor prognosis. Isolated organ (bone, kidney, adrenal, meninges; spine = Pott's): good prognosis with treatment.
- Complications: pleural infection, intestinal TB from swallowed sputum, haemoptysis, extensive fibrosis → heart failure.

Immunosuppressed host

- Diffuse caseation or miliary spread instead of fibrosis.
- MAC (atypical, opportunistic): AIDS with low CD4; lungs and GI tract; fever, drenching sweats, weight loss, diarrhoea, RUQ pain; water/hot tubs/soil. Macrophages full of AFB, granulomas and necrosis rare.

Source: Pulmonary tuberculosis [010] (full and short decks)

PART 1: LECTURE

Pathology of infectious diseases [VM 113]

Topic: Pathology

Pneumonia basics

- Patchy or diffuse alveolar inflammation with consolidation. Types by site: lobar, lobular (bronchopneumonia), interstitial.
- Risks: immune defects (neutrophil/complement/humoral → bacterial; cell-mediated → mycobacteria), suppressed cough (coma, anaesthesia), smoking, obstruction, oedema, chronic disease, malignancy.

Lobar pneumonia vs bronchopneumonia

- Lobar: acute diffuse fibrinous inflammation of one or more lobes; adults; pneumococci by droplets; DM and malignancy predispose.
- Stages: congestion (1-2 d) → red hepatization (3-4 d: red, firm, liver-like; neutrophils, RBCs, fibrin) → grey hepatization (~1 wk: grey-brown, dry; RBCs disintegrate, more fibrin, clear zone by alveolar wall) → resolution (1-2 wk).
- Complications: organization (firm lung), abscess, septicaemia (endocarditis, pericarditis, meningitis, arthritis), toxic myocarditis/heart failure.
- Bronchopneumonia: patchy, bilateral, basal; extremes of age; hospital acquired (Staph aureus, Klebsiella, E. coli, Pseudomonas); nodules around bronchioles exuding pus; lysis and complications common.

Lung abscess

- Suppurative (liquefactive) inflammation; late abscess: necrotic pus centre, pyogenic membrane, granulation tissue.
- Acute (4-6 wk) vs chronic (> 6 wk); primary vs secondary.
- Main mechanism: aspiration with mouth anaerobes (periodontal disease, seizures, alcohol, dysphagia, reduced consciousness). Others: septic emboli (often multiple) from bacteraemia or tricuspid endocarditis, infected bulla/cyst, obstruction by bronchial carcinoma.
- Microbiology rarely confirmed: samples contaminated by upper airway flora.
- 90% cure with antibiotics; poor outlook in elderly, HIV, immunosuppression, malignancy, symptoms > 8 wk; ~75% mortality if immunocompromised or obstructed.

Atypical and opportunistic

- Atypical: septal inflammation, no consolidation; wide septa with lymphocytes, histiocytes, plasma cells; empty alveoli. Mycoplasma, Legionella, Chlamydia, Coxiella, Pneumocystis, viruses. Mild; may heal by fibrosis.
- Opportunistic: CMV, measles, Pneumocystis jirovecii (commonest HIV opportunistic infection; foamy alveolar exudate), Aspergillus, Candida.

Aspergillosis (branching septate hyphae)

- Simple aspergilloma: fungus ball in old cavity (TB, cyst), no invasion, immunocompetent.
- Chronic: existing cavity + lowered defences; hyphae invade cavity wall, inflammation, fibrosis, necrosis.
- Invasive: immunocompromised, days-weeks, angioinvasion; nodule of coagulative necrosis with radial hyphae and haemorrhagic rim, or lobular bronchopneumonia.
- ABPA: hypersensitivity to inhaled spores.

Source: Pathology of infectious diseases [VM 113]

PART 1: LECTURE

Pathology of lung tumours [VM 011, 067]

Topic: Pathology

Lung cancer basics

- Bronchogenic carcinoma = 95% of primary lung tumours; peak 50-70 y; most common fatal cancer.
- Risks: smoking (pack-years; 80% of cancers), uranium, asbestos, arsenic, nickel, radiation, radon, air pollution, genetics, scars.
- Late presentation: cough, dyspnoea, weight loss, chest pain, metastases (adrenal, brain, liver, bone), paraneoplastic syndromes.
- Pancoast (superior sulcus): shoulder/ulnar arm pain, hand wasting, Horner syndrome, oedema, hoarseness; mostly SCC or adenocarcinoma.

NSCLC

- SCC (30%, M>F, smokers): central, main bronchus; metaplasia → dysplasia → CIS; keratin pearls, intercellular bridges; hypercalcaemia (PTH-rp); HMW CK+, TTF-1−.
- Adenocarcinoma (30%, F>M, commonest in female non-smokers): peripheral, scar cancers; glands + mucin; clubbing, hypercoagulability; TTF-1+, HMW CK−. Sequence: atypical adenomatous hyperplasia (K-RAS) → adenocarcinoma in situ (≤ 3 cm, along alveolar walls, no invasion) → invasive.
- Large cell (10%): undifferentiated, high grade, poor prognosis, paraneoplastic rare.
- Treatment: surgery if resectable without metastasis; radiotherapy if not resectable; chemo if metastatic (poor response).

Small cell (SCLC)

- Neuroendocrine; 10-20%; 99% smokers; central; small cells, scant cytoplasm, nuclear moulding, high mitoses, necrosis; chromogranin+, dense-core granules.
- ACTH (Cushing), ADH (SIADH). Usually metastatic at diagnosis. Chemo (initially very sensitive), radiotherapy for SVC obstruction, surgery contraindicated. Staged limited vs extensive.

TNM (NSCLC)

- T1 ≤ 3 cm; T2 > 3-5 cm or main bronchus, visceral pleura, atelectasis; T3 > 5-7 cm or chest wall, phrenic, parietal pericardium; T4 > 7 cm or mediastinum, heart, great vessels, carina, etc.
- N1 ipsilateral hilar; N2 ipsilateral mediastinal/subcarinal; N3 contralateral, scalene, supraclavicular. M1a pleural/contralateral lung; M1b single, M1c multiple extrathoracic.

Carcinoid, metastases, pleura, diagnosis

- Carcinoid: < 40 y, M=F, no smoking link; central polypoid endobronchial mass; nests, salt-and-pepper chromatin; atypical = more mitoses + necrosis. Serotonin → flushing, diarrhoea, wheeze, tricuspid/pulmonary valve scarring.
- Metastases: multiple well-defined nodules, usually adenocarcinoma; TTF-1+ = lung (or thyroid), TTF-1− = secondary.
- Mesothelioma: asbestos (latency up to 40 y; smoking no effect); plaques → pleural encasement; epithelioid (best), sarcomatous, biphasic; calretinin+, TTF-1−; 50% die within 1 y.
- Sampling: central → bronchoscopy; peripheral → CT-guided FNA/biopsy; pleural → fluid + biopsy.

Source: Lung tumours lecture; Lung tumours revision

PART 1: LECTURE

Basic chest X-ray interpretation [VM 080]

Topic: Radiology

Routine and technique

- Systematic routine: airways, lungs, heart, hila, bones and soft tissues, hidden areas.
- Check technique first: AP film → heart and mediastinum look enlarged; poor inspiration → large heart, hazy lungs; underpenetrated → too white; overpenetrated → too dark; rotated → one side denser.
- Silhouette sign: an opacity touching the heart, aorta or diaphragm hides that border. Right heart border = right middle lobe; left heart border = lingula; left lower lobe pneumonia leaves the left heart border visible.

Too white

- Consolidation (pus, blood, water, cells, protein): classic sign = air bronchogram; fissure-bounded lobar opacity.
- Interstitial: reticular, coarse thickened interstitium (fibrosis, asbestosis, sarcoid, radiotherapy, drugs).
- Collapse: opacity + volume loss + shift of trachea, mediastinum, hilum or fissure towards it (ET tube, tumour, mucus plug, foreign body). Right upper lobe: horizontal fissure rises. Left lower lobe: sail-shaped retrocardiac opacity.
- Focal lesion/coin: cancer (spiculated mass), metastasis, hamartoma, pneumonia, abscess, infarct.
- Cavity: dark centre ± air-fluid level (TB, Staph, tumour, infarct, collagen-vascular disease).
- Bronchiectasis: tram tracks (cylindrical), air-fluid levels (cystic); CXR not adequate to diagnose.
- Multiple nodules: > 5 mm mostly metastases; < 5 mm metastases, sarcoid (with reticulonodular shadowing and hilar nodes), miliary TB, pneumoconiosis, chickenpox.
- Pleura: effusion blunts costophrenic angle with meniscus rising to axilla; plaques/calcification (asbestos, TB); mass (mesothelioma, metastasis, fibroma).

Too black

- COPD: hyperexpansion, flat diaphragms, wide rib spaces, bullae, but lung markings still present.
- Pneumothorax: lung edge with no markings beyond it.
- Tension pneumothorax: emergency; large pneumothorax, collapsed lung, trachea/mediastinum shifted away, flat diaphragm.

Heart, hila, hidden areas

- Cardiomegaly: heart diameter > 50% of the transverse (thoracic) diameter; heart failure, pericardial effusion, valve disease.
- Heart failure ABCDE: alveolar oedema (bat wing), Kerley B lines, cardiomegaly, dilated upper lobe vessels, effusion.
- Hila: enlarged (bilateral lymphadenopathy, pulmonary hypertension; unilateral → primary carcinoma), too dense, or displaced (collapse).
- Hidden areas: apices, behind the heart (collapse, hiatus hernia with air-fluid level), under the diaphragm (free air, subphrenic collection).

Source: Basic chest X-ray interpretation [VM 080]

PART 1: LECTURE

Radiology of TB [VM 085]

Topic: Radiology

Course of pulmonary TB

- Primary complex → parenchymal and nodal spread → miliary TB (blood spread, widespread granulomas) → healing with fibrosis and calcification → reactivation.
- Past surgical treatment: artificial collapse of the lung.

Primary TB

- Parenchymal consolidation, lymphadenopathy, pleural effusion, Ghon complex.
- Ghon focus may appear as a dense nodular opacity (e.g. right mid-zone) with hilar nodes.
- A small focus may be invisible on CXR; CT can show it (e.g. posterior segment, right upper lobe).
- Pleural effusion: homogeneous opacity rising to the axilla.

Post-primary TB

- Patchy upper-zone consolidation, cavitation, then fibrosis; pleural disease.
- CT: upper lobe cavity, atelectasis/fibrotic bands, traction bronchiectasis.
- Healed: bilateral apical opacities with fibrosis, volume loss, upward retraction, calcific foci.
- Suspect in chronic cough not helped by antibiotics, night sweats, haemoptysis.

Miliary and extrapulmonary TB

- Miliary: innumerable 1-3 mm nodules throughout both lungs (haematogenous).
- Pericardial effusion: enlarged flask-shaped heart.
- Brain tuberculomata.
- Pott's disease: vertebral TB.

At-risk patients and old treatments (lecture cases)

- Migrant children from high-prevalence countries, homeless people, former IV drug users and those on treatment for other illness, elderly patients with past TB.
- Plombage: inert material (Lucite balls, rubber, paraffin wax, gauze) placed under the upper ribs to collapse the upper lobe, as a collapsed lung was thought to heal faster; seen as balls in the upper pleural space.

Source: Radiology TB [VM 085]

PART 1: LECTURE

Arterial imaging [VM 015]

Topic: Radiology

Modalities

- Doppler ultrasound: carotid, upper limb and lower limb arterial studies; non-invasive, no radiation.
- Angiography options: conventional catheter angiography (invasive), CT angiography (CTA) and MR angiography (MRA).
- CTA is non-invasive and fast compared with conventional catheter angiography, but uses X-rays and iodinated contrast.
- MRA gives a vessel map without ionising radiation.

Reading an arterial occlusion

- Look for: site of block, atheromatous calcification (aorta, iliac, leg arteries), collaterals, distal run-off.
- Good collaterals refilling the popliteal artery with good distal run-off = chronic occlusion.
- Poor or absent collaterals with weak distal filling = suggests acute occlusion.
- MRA of leg: tibial artery occlusion with multiple collateral branches = chronic occlusion.
- Bilateral SFA occlusion may coexist with aorto-iliac calcification and a small femoral artery aneurysm.

Anatomy to know for the cases

- SFA occlusion is bypassed mainly by profunda femoris branches linking with popliteal genicular arteries.
- Popliteal artery ends as anterior tibial and posterior tibial (tibioperoneal trunk) arteries; gives genicular branches.

Aneurysms

- Fusiform = circumferential spindle-shaped dilatation (e.g. abdominal aorta, descending thoracic aorta).
- Saccular = one-sided sac-like outpouching (case: infrarenal abdominal aorta).

Intervention

- SFA occlusion: interventional treatment = percutaneous transluminal angioplasty (PTA) and stenting.

Source: Arterial vascular imaging [VM 015] lecture and computer lab

PART 1: LECTURE

Venous imaging [VM 024]

Topic: Radiology

Modalities for DVT

- Duplex/colour Doppler with compression, contrast venography, CT venography, MR venography, impedance plethysmography.
- Duplex ultrasound = investigation of choice for diagnosis and follow-up of acute DVT: non-invasive, painless, available.
- Highly sensitive for occlusive proximal thrombi, less so for calf thrombi. Most lower limb DVTs begin in the soleal veins of the calf.

Duplex findings in DVT

- Non-compressible venous segment (key sign); absent colour flow if fully occlusive.
- Loss of phasic flow on Valsalva; no flow augmentation with calf squeeze; increased flow in superficial veins.
- Acute thrombus: increased vein diameter, anechoic (dark) thrombus.
- Chronic thrombus: decreased vein diameter, echogenic (bright) thrombus.

Doppler technique

- Spontaneity: flow without augmentation. Phasicity: flow changes with respiration.
- Compression: in transverse plane. Augmentation: squeeze distal to probe = patency below the probe.
- Valsalva: deep breath and strain = patency of abdominal and pelvic veins.

Other modalities

- Contrast venography: historical gold standard, now replaced by duplex; invasive, expensive, painful; thrombus = filling defect; most useful in non-post-phlebitic legs.
- CT venography: for veins above the common femoral vein (IVC, iliac) where ultrasound is less accurate; lecture = modality of choice for post-phlebitic limb and deep vein patency.
- MR venography: useful but time, cost, resources; signs include filling defects, vein wall thickening, perivascular oedema.
- Impedance plethysmography: good sensitivity for proximal DVT, poor specificity.

Source: Venous imaging [VM 024] lecture and computer lab

PART 1: LECTURE

Imaging of thromboembolic disease and PE [VM 091]

Topic: Radiology

CT pulmonary angiography (CTPA)

- First-choice test for PE; negative CTPA = no further investigation or treatment for PE.
- Best with underlying cardiorespiratory disease; can confirm an alternative diagnosis.
- PE = filling defect in a contrast-filled pulmonary artery (main, lobar or segmental).
- Pregnancy test first in women of child-bearing age. Not appropriate if D-dimer negative and Wells says PE unlikely.
- May rarely miss a small peripheral PE.

Alternatives

- V/Q scan: good first-line alternative; much lower radiation. Preferred if age < 40, contrast allergy or severe renal impairment.
- PE on V/Q = perfusion defect with normal ventilation. Normal V/Q virtually excludes PE.
- V/Q hard to interpret if CXR abnormal or cardiopulmonary disease (e.g. COPD): use CTPA instead.
- Conventional pulmonary angiography: most accurate but invasive.
- MR pulmonary angiography: contrast-enhanced = no radiation but contrast hazards; non-contrast cardiac-gated = no radiation or contrast, alternative in renal impairment and pregnancy, less accurate than CT.

Supporting tests

- Chest X-ray: often normal; wedge-shaped peripheral opacity (Hampton's hump), small effusions, oligaemia in massive PE; excludes pneumonia, pneumothorax, heart failure.
- Echocardiography: reliably diagnoses massive PE (acute right heart strain or failure).
- Leg Doppler: most PE patients also have DVT; use when probability high (Wells > 4 or positive D-dimer) but CTPA negative.

Treatment link

- IVC filter is life saving in massive DVT to prevent PE; filters can be placed and later removed.

Source: Imaging of thromboembolic disease (PE) [VM 091]

PART 1: LECTURE

ICC-1 spotting: anatomy and radiology

Topic: Radiology

Exam format

- 40 marks: Anatomy.TV video (20 marks, 5-point answer, e.g. artery = beginning, end, branches, surface anatomy, pulsation site) + 10 stations (2 min, 2 marks each).

Thorax anatomy

- Right lung 10 segments: upper 3 (apical, posterior, anterior), middle 2 (lateral, medial), lower 5. Left lung 8: upper 4 (apicoposterior, anterior, superior and inferior lingular), lower 4.
- Bronchial arteries: right from 3rd right posterior intercostal; left from descending thoracic aorta. Bronchial veins: right to azygos, left to superior hemiazygos.
- SVC = union of right and left brachiocephalic veins. Right atrium receives SVC, IVC, coronary sinus.
- Arch of aorta branches: brachiocephalic, left common carotid, left subclavian. Concavity: left bronchus, ligamentum arteriosum, superficial cardiac plexus.
- Ascending aorta gives right and left coronaries. RCA: marginal, posterior interventricular. Apex: left 5th ICS, 9 cm from midline.

Abdomen and limbs

- Coeliac trunk: left gastric, common hepatic, splenic. SMA: inferior pancreaticoduodenal, middle colic, right colic, ileocolic. IMA: left colic, sigmoid, continues as superior rectal.
- Femoral artery: superficial circumflex iliac, superficial external pudendal, superficial epigastric, profunda femoris, descending genicular.

Radiology spots

- TB: Ghon focus; cavity with hilar node. PE: filling defect in pulmonary artery; saddle embolus; poor contrast filling of a zone.
- Arterial: SFA occlusion with collaterals = chronic; poor collaterals = acute; fusiform and saccular aortic aneurysms.
- Venous: compression on Doppler excludes DVT; no colour fill + thrombus = thrombophlebitis; filling defect on venogram = thrombus.

Chest X-ray (ABCDE)

- Adequate PA film: > 6 anterior or > 10 posterior ribs above diaphragm; spinous processes midway between clavicles.
- Effusion: blunted or obliterated costophrenic angle. COPD: flat diaphragms, wide rib spaces, lucency. Pneumothorax: pleural edge, no markings.
- CTR normal ≤ 0.5; mediastinum < 8 cm (wider = think dissection). Right hemidiaphragm higher by up to 1 ICS.

Source: Year 3 ICC-1 Spotting Final

PART 1: LECTURE

Management of disturbed behaviour [VM 031] (lecture and LGW)

Topic: Mental health

Causes and risk factors

- Model: Antecedents → Behaviour → Consequences; aim = prevention → management → harm reduction.
- Organic: unmet needs, pain, delirium/physical illness, head injury, epilepsy, dementia, intoxication (alcohol, cocaine, amphetamines, MDMA, LSD; fluctuates hour by hour; urine drug screen helps) and withdrawal.
- Delirium tremens (alcohol withdrawal): clouded consciousness, disorientation, visual/auditory/tactile hallucinations; medical emergency, mortality 5-10%.
- Psychiatric high-risk symptoms: command hallucinations, delusions/hallucinations focused on a person, violent delusions of control, violent fantasy. Mania is a psychiatric emergency. Antisocial and emotionally unstable personality disorders.

Prediction and first steps

- Warning signs: tense angry face, prolonged restlessness, louder speech, prolonged eye contact, violent psychotic content, blocking escape routes, verbal threats.
- Principles: ensure safety, rule out organic cause, act swiftly, confidentiality does not override serious risk, work with seniors, keep records, "do no harm".
- If delirium suspected → treat underlying cause (e.g. post-op fever, confusion).

Hierarchy of interventions (proportionate to risk)

- Verbal de-escalation / talking down, low-stimulus room → time out → rapid tranquilisation → physical restraint (trained staff only) → seclusion.
- Time out: max 15 min, never a locked room. Seclusion: last resort, never a planned treatment, observe constantly, review every 2 h.
- Leaving A&E: cannot use Section 5(2); hold under Mental Capacity Act if lacks capacity.

Rapid tranquilisation (last resort)

- Before: legal status, history/MSE/exam, flumazenil available. Offer oral before IM.
- Lorazepam 1-2 mg (max 4 mg/24 h) ± haloperidol 5-10 mg (max 30 mg/24 h PO, 18 mg/24 h IM). Non-psychotic: lorazepam; psychotic: lorazepam + antipsychotic; lower doses in elderly.
- After: T, P, BP, RR every 5-10 min for 1 h; ECG formally recommended for all given haloperidol.

Adverse effects

- Akathisia: inner restlessness, pacing → reduce/change drug, propranolol, clonazepam, mirtazapine.
- Acute dystonia (oculogyric crisis, torticollis; young neuroleptic-naive males, haloperidol) → procyclidine.
- NMS: fever, rigidity, altered mental state, autonomic instability, ↑↑CK → stop drug, supportive care, bromocriptine + dantrolene; restart after ≥5 days with an unrelated drug.
- Respiratory depression (RR <10 or SpO2 <90%) → oxygen, flumazenil if benzodiazepine.
- QTc >500 ms → arrhythmia risk; high risk: IV antipsychotics, haloperidol, methadone; worsened by low K, Mg, Ca.

Source: VM-MH017.1 Management of disturbed behaviour lecture; VM-MH017.2 LGW tutor/student notes

PART 1: LECTURE

Depression, anxiety and obsession [VM 042]

Topic: Mental health

Session focus

- Three "neurotic" disorders: depression, anxiety, obsessive-compulsive disorder (OCD).
- For each know: main symptoms, diagnostic criteria, and biological, psychological and social treatment.
- Key references: NICE guidance on depression, anxiety and OCD.

Depression (case: 67-year-old woman 4 weeks after CABG)

- Clues: tearful, withdrawn, not eating, awake at night, not engaging in rehab; normal observations.
- Core symptoms: low mood, loss of interest/enjoyment, low energy; plus poor sleep, appetite, concentration.
- Common after cardiac surgery and slows recovery; seek liaison psychiatry advice.
- Drug choice after a cardiac event: sertraline (SSRI); avoid tricyclics. On antiplatelets/ anticoagulant → watch bleeding (PPI helps).
- Look at social factors (husband berating her).

Anxiety (case: 30-year-old woman collapsing in crowded shops)

- Panic attacks: sudden fear of dying, racing heart, breathlessness, sweating, dry mouth; repeated → panic disorder; in crowded places → agoraphobic features.
- Daily worry from waking for >2 months → generalised anxiety.
- Exclude physical causes first: ECG, bloods, CXR normal.
- Treatment: CBT; if medication needed → SSRI; avoid long-term benzodiazepines.

OCD (case: 20-year-old student washing hands in bleach)

- Contamination obsessions + washing compulsions; time-consuming, impair study and social life.
- Red cracked skin = damage from washing, not infection → antibiotics not indicated.
- Treatment: CBT with exposure and response prevention ± SSRI.
- Address coping with alcohol and cannabis.

Measurement scales

- BDI: self-rated depression inventory.
- HADS: self-rated anxiety and depression in hospital/medically ill patients.
- Hamilton Depression Scale: clinician-rated.

Source: VM-MH018.1 Depression Anxiety and Obsession student notes (case vignettes; answers are standard NICE-based teaching, the notes give questions only)

PART 1: LECTURE

Drug misuse presenting to A&E [VM 075]

Topic: Mental health

Opiates (heroin)

- Overdose: pinned pupils, shallow breathing, cyanosis, low SpO2, loud snoring, unrousable; often mixed with benzodiazepines/alcohol.
- Treat with naloxone; short half-life → may become unconscious again if he leaves. Flumazenil reverses benzodiazepines.
- Withdrawal: sweating, goose-bumps, dilated pupils, runny nose, watery eyes, yawning, cramps, diarrhoea, tachycardia.
- In A&E: no methadone usually; symptomatic (diazepam, loperamide, buscopan). On AMU: methadone/buprenorphine per protocol; confirm prescription with agency and pharmacy; urine drug screen; refer to drug service early.

Other drugs

- Cocaine/crack: stimulant, binge use; hypertension, fits, MI, stroke, gut infarction, rhabdomyolysis; snorting → perforated septum.
- GHB/GBL: GABA-B, sedation, respiratory depression (worse with alcohol); narrow overdose margin. GBL withdrawal → delirium, hallucinations; treat with benzodiazepines + baclofen, may need ITU.
- MDMA: dilated pupils, jaw clenching, ↑temperature; deaths from heatstroke, dilutional hyponatraemia, heart failure.
- Benzodiazepine withdrawal: anxiety, depersonalisation, light/sound sensitivity, fits → diazepam detox.
- Injecting: abscess, cellulitis, septicaemia, acute endocarditis, DVT/PE (groin), arterial injection ischaemia; HIV, hepatitis B/C.

Diagnosis (ICD-10)

- Harmful use: use causing physical or mental damage (e.g. low mood after crack binge).
- Dependence: craving, poor control, use despite harm, priority over other activities, tolerance, ± withdrawal.

Alcohol

- Withdrawal: sweating, tremor, tachycardia, anxiety, vomiting; fits; delirium tremens (disorientation, visual hallucinations) → consider haloperidol.
- Wernicke: confusion, ataxia, nystagmus, ophthalmoplegia → IV thiamine (correct Mg, glucose). Korsakoff: loss of new learning/short-term memory.
- AMU: SADQ (<16 mild, 16-30 moderate, >30 severe); chlordiazepoxide reducing regimen; thiamine/Pabrinex; seizures → rectal diazepam 10 mg.
- A&E intoxication: ABCDE, risk of aspirating vomit, check glucose. GP screening: AUDIT/AUDIT-C, CAGE; brief interventions.

Safe prescribing and drug seeking

- Red flags: requests for specific opioids, vague pain, doctor shopping, "lost" methadone. But may be genuinely ill.
- Avoid unreviewed repeats; use instalment, small, time-limited prescriptions.

Source: VM-MH019.1 Presentations of drug misuse to A&E lecture; VM-MH019.1 Overview of illicit drugs

PART 1: LECTURE

Egyptian Mental Health Act and involuntary management [VM 084]

Topic: Mental health

Capacity

- Ability to understand, retain, weigh information and communicate a decision.
- Presumed unless proved otherwise; task- and time-specific; support decision-making; unwise decision ≠ lack of capacity; best interests and least restrictive option.
- Help capacity: no jargon, pictures, translators, treat pain, time, quiet setting, relative present.
- Loss may be temporary (unconscious), permanent after having it (dementia) or lifelong (severe learning disability).

The Act (2009, presidential decree)

- Tight criteria for detention; court review of detention; notify public prosecutor within 24 h of involuntary admission.
- Voluntary patients consent to treatment; documented, reviewed treatment plans; patients' rights committee in each facility; sanctions for violations.

Admission articles

- Art 10: anyone 18+ can request voluntary admission and discharge (unless involuntary criteria apply).
- Art 11: voluntary patient may be held ≤72 h if serious imminent danger or cannot self-care; only emergency treatment without consent; extendable ≤1 week.
- Art 13: involuntary admission needs a psychiatrist's approval, severe illness, refusal, plus either imminent severe deterioration or serious imminent threat to self/others; notify within 24 h.
- Art 14: non-psychiatrist may admit for assessment ≤48 h on written request (relative to 2nd degree, police, social worker, health inspector, consul, unrelated outside psychiatrist); prosecutor within 24 h.
- Art 15: extend to max 7 days if independent assessment not possible.
- Art 16: >1 week needs two registered psychiatrists' assessments sent to Regional Council within 7 days, or detention ends.
- Art 21: escaped patient → police/prosecution return him.

Treatment articles

- Art 28: refusing involuntary patient can be compelled; review ≤ every 4 weeks; independent assessment if >3 months.
- Art 29: emergency treatment without consent ≤72 h.
- Art 30: ECT only under general anaesthesia + muscle relaxant with written informed consent; refusing involuntary patient needs independent evaluation.

Seclusion, restraint, observation

- Last resort, shortest time, not therapeutic. Restraint: physical, mechanical or chemical.
- Constant (within reach): extremely suicidal. Level 1: every 15 min, stays on unit. Level 2: every 30 min. Level 3: hourly whereabouts.

Source: VM-MH020.1 Egyptian Mental Health Act Involuntary management

PART 1: LECTURE

Obsessive compulsive disorder [VM 088]

Topic: Mental health

Definition (ICD-11 / DSM-5)

- Obsessions: repetitive, persistent intrusive and unwanted thoughts, images or urges (contamination, violent scenes, urge to stab), usually with anxiety; person tries to ignore, suppress or neutralise them.
- Compulsions: repetitive behaviours (washing, checking, ordering) or mental acts (counting, praying, repeating words) done in response to an obsession to reduce anxiety.
- Time-consuming (e.g. >1 h/day) or significant distress or impairment.

Key features

- Obsessions are distressing (often violent or obscene) and recognised as the person's own thoughts (unlike psychosis).
- Compulsions are not enjoyable, do not complete useful tasks, and are meant to prevent an unlikely event.
- Cycle: obsession → anxiety → compulsion → brief relief → repeats.
- Distinguish from anankastic (obsessional) personality traits.

Epidemiology

- Adult prevalence 0.6-2.5%; higher in children.
- Mean onset age 26; M:F 1:1, men present earlier.

Aetiology

- Genetic: first-degree relative → 5-7× risk; up to 80% concordance in identical twins.
- Hyperactive orbitofrontal cortex, anterior cingulate cortex, caudate nucleus; impaired inhibitory response.
- Serotonergic pathways; possible immunological cause (PANDAS: after streptococcal infection in children).

Treatment

- Psycho-education: explain symptoms, simple strategies, signpost support groups.
- Psychotherapy: CBT (main), metacognitive, mentalisation-based, psychodynamic.
- Medication: SSRI (e.g. sertraline, often high dose); clomipramine (tricyclic) if SSRI fails.
- Course can relapse; resistant cases may need long-term therapy or specialist inpatient programmes.

Source: VM-MH021.1 Obsessive Compulsive Disorder (OCD)

PART 1: LECTURE

Reactions to stress including PTSD [VM 089]

Topic: Mental health

Acute stress reaction (ICD-11)

- Follows an extremely threatening or horrific event; ICD-11 does not count it as a mental disorder.
- Emotional, somatic (daze), cognitive (amnesia, depersonalisation, stupor) and behavioural symptoms; autonomic signs (tachycardia, sweating, flushing) common.
- Starts within hours to days; settles within a few days or once removed from the threat.
- Manage: psychoeducation, normalise symptoms, practical support. TCAs, benzodiazepines, anxiolytics: little evidence.

Acute stress disorder (DSM-5) and adjustment disorder

- ASD: actual or threatened death, serious injury or sexual violence + intrusion, negative mood, dissociation, avoidance, arousal; lasts 3 days to 1 month.
- Adjustment disorder: maladaptive reaction to a psychosocial stressor; preoccupation, worry, rumination; impaired functioning.
- Onset usually within 1 month; resolves within 6 months once stressor and consequences end.

PTSD: stressor and features

- DSM-5 exposure: directly experiencing, witnessing in person, learning it happened to a close relative or friend (violent or accidental), repeated exposure to details (first responders).
- Causes: combat, road accidents, terrorism, disasters, hostage-taking, assault.
- 4 core groups: intrusion (flashbacks, nightmares), avoidance, negative cognition and mood (guilt, detachment), arousal (startle, poor sleep and concentration). Duration > 1 month.
- Lifetime prevalence 7.8%; F > M.
- Comorbidity: affective and anxiety disorders, substance misuse, somatisation; family problems, violence; poor physical health.

Aetiology and risk

- Psychological: fear conditioning, appraisal, traumatic memories, maintaining behaviours.
- Biological: raised adrenaline and noradrenaline; low cortisol with HPA hypersuppression; genetic.
- Person: female, Black or Hispanic, family history, previous trauma, low intelligence or self-esteem.
- Stressor: prolonged or repeated, grotesque, intentional, harm to children, loss of function (not injury severity).

Treatment of PTSD

- Psychological: CBT, EMDR, psychodynamic therapy, hypnotherapy.
- Drug: SSRIs. Medication does have a role.

Source: VM 089 Reactions to stress including PTSD

PART 1: LECTURE

Care of the critically ill patient [VM 003]

Topic: Mental health

Recognition and principles

- Most arrests are predictable: deterioration comes before 50-80% of cardiac arrests.
- Hypoxia and hypotension are common warning signs; delays in referral to higher care add to the problem.
- Early warning scores with an escalation protocol detect ward deterioration.
- Complete the initial assessment, treat life-threatening problems as found, reassess, check the effect of each treatment, call for help early (assess, treat, reassess).
- Initial: personal safety, responsiveness, first impression (movement, pulse, breathing effort, temperature).

A: Airway

- Causes: CNS depression, blood, vomit, foreign body, trauma, infection, inflammation, laryngospasm, bronchospasm.
- Signs: noisy breathing (stridor, wheeze, gurgling), choking, see-saw breathing, accessory muscles. A talking patient has an open airway.
- Treat: head tilt, chin lift, jaw thrust; simple adjuncts; advanced (LMA, tracheal tube); oxygen; early capnography.

B: Breathing

- Causes: reduced drive (CNS depression); reduced effort (weakness, nerve damage, chest defect, rib pain); lung disease (pneumothorax, haemothorax, infection, COPD, asthma, PE, ARDS).
- Look (distress, cyanosis, rate), listen (breath sounds), feel (expansion, percussion, tracheal position).
- Treat: oxygen, treat cause (drain pneumothorax), bag-mask if inadequate, monitor SpO2 and rate.

C: Circulation

- Primary (cardiac): ACS, arrhythmia, hypertensive heart disease, valve and hereditary disease. Secondary: asphyxia, hypoxaemia, blood loss, hypothermia, septic shock.
- Assess pulse, capillary refill, BP, organ perfusion (chest pain, mental state, urine output), bleeding; record ECG.
- Treat: IV/IO access and bloods, ECG monitoring, fluid challenge, treat cause, inotropes or vasopressors; aspirin, nitrates, oxygen (if needed), morphine for ACS.

D and E

- D: AVPU or GCS, pupils, lateralising signs; glucose < 4 mmol/L give glucose; lateral position; check drug chart.
- E: undress to examine (injuries, bleeding, rashes), avoid heat loss, keep dignity.

Source: VM 003 Care of critically ill patient

PART 1: LECTURE

People with learning disabilities [VM 014]

Topic: Mental health

Definition and causes

- Criteria: impaired intellectual function + impaired adaptive function + onset in the developmental period.
- DSM-5: IQ < 70 indicates LD; level (mild to profound) set by adaptive functioning.
- Prevalence: mild 1-2%, severe and profound about 0.35%.
- Causes: idiopathic (often), genetic, substance misuse (foetal alcohol syndrome, preventable), perinatal, environmental.

Key syndromes

- Down's: trisomy 21 (also translocation, mosaic); maternal-age related; AVSD, hypothyroidism, leukaemia, coeliac, cataracts; dementia at a younger age, depression.
- Fragile X: commonest inherited LD in boys and single-gene cause of autism; CGG repeats on distal Xq; hyperactivity, social and speech deficits; tremor/ataxia in older carrier males; premature ovarian failure in carrier females; worsens through generations.

Physical health

- More illness, poor access to care, late presentation; 37% of deaths avoidable in one inquiry.
- Epilepsy 22% vs 1%; dementia 21.6% vs 5.7%; schizophrenia 3% vs 1%; hearing problems 40%; thyroid problems.

Mental illness

- Dual diagnosis common; change in behaviour is highly significant.
- Diagnostic overshadowing: blaming new symptoms on the LD. Exclude physical causes (look for pain) first.
- Depression: often missed; atypical (hypersomnia, increased appetite), self-injury or aggression.
- Schizophrenia: more in mild LD; velo-cardio-facial syndrome; simple delusions.
- Challenging behaviour is not a diagnosis; multifactorial.

Assessment and management

- History from several sources; observe in natural settings; serial examinations.
- Bio-psycho-social, multidisciplinary, use mainstream services, assess capacity.
- Drugs: treat illness; 2nd line for behaviour; start low, go slow; never only to sedate; physical check, ECG where indicated, bloods; review at 2 weeks.
- GMC: extra time, simple words or pictures, demonstrate procedures first, reasonable adjustments.
- Safeguarding: watch for abuse and financial exploitation.

Source: VM 014 People with learning disabilities

PART 1: LECTURE

Self-harm and suicide [VM 065]

Topic: Mental health

Self-harm: definition and epidemiology

- NICE: self-poisoning or self-injury, irrespective of apparent purpose. 'Deliberate self-harm' is outdated and stigmatising.
- Lifetime prevalence about 5% (13% in adolescents); much is hidden.
- Risk factors: adolescence (peak women 15-24, men 25-34), female, separated or divorced, low income, abuse, isolation, alcohol (half), prisoners, early psychosis.
- 90% have a psychiatric disorder: depression (2/3), substance misuse, anxiety; personality disorder in half.
- Lithium (bipolar) and clozapine (resistant schizophrenia) reduce self-harm and suicide; SSRI effect controversial.

Methods and paracetamol

- Self-poisoning ~90% (paracetamol, antidepressants, benzodiazepines, often with alcohol); self-injury ~10%.
- Paracetamol: commonest cause of acute liver injury; 10-15 g severe, 5 g if high risk. NAPQI depletes glutathione.
- Activated charcoal if serious overdose within 2 h; N-acetylcysteine is a glutathione precursor.

Assessment

- Everyone needs assessment of needs and risk, in parallel with medical care; refer to liaison psychiatry. Be non-judgemental.
- Intent: precautions against discovery, note or affairs in order, planning, disclosure, regret. Physical severity does not show intent.
- Get collateral history; risk scales not recommended.
- Self-harm raises suicide risk 50-100 times; highest within 6 months.
- Suicide risk after self-harm: older, male, isolated, mental or physical illness, hopelessness, substance misuse, violent method, strong intent.

Risk level and plan

- Low (no intent, impulsive, supportive, no history): back to GP.
- Medium (frequent thoughts, previous attempts, low support): follow-up, contingency plan, crisis team?
- High (clear plan, major depression, violent method, regret it failed): crisis team, admission, Mental Health Act assessment.

Suicide

- About 1 million deaths a year; M:W 4:1 in the West; most were depressed.
- Factors: family history, mental illness, pain or terminal illness, alcohol or drugs (2/3), access to means (doctors), prisoners.
- Asking about suicide does not cause it. Clusters occur: do not publicise methods.

Source: VM 065 Self harm and suicide

PART 1: LECTURE

Retinal vascular diseases [VM 043]

Topic: Ophthalmology

Diabetic retinopathy

- Commonest cause of blindness in working-age adults. Microangiopathy: loss of pericytes, microaneurysms, capillary closure, ischaemia, leakage.
- Risk: duration, poor control, hypertension, renal disease.
- NPDR: microaneurysms, dot and blot haemorrhages, hard exudates.
- Macular oedema: commonest cause of visual loss in NPDR; any stage.
- Severe NPDR: cotton wool spots (nerve fibre layer infarcts), venous beading, IRMA.
- PDR: ischaemic retina releases VEGF; NVD/NVE. Advanced: vitreous haemorrhage, tractional detachment, neovascular glaucoma.
- Treat: glucose control; anti-VEGF for oedema; PRP for severe NPDR (high risk) and PDR (side effects: peripheral and night vision loss); vitrectomy for advanced disease.
- Silent until late: annual retinal check for all diabetics.

Hypertensive retinopathy

- Grade 1 barely detectable arterial narrowing; 2 obvious narrowing with focal irregularity; 3 = 2 + haemorrhages +/- exudates; 4 = 3 + disc swelling.

Retinal vascular occlusion

- Both: sudden painless loss of vision; check BP, lipids, glucose, FBC, ESR/CRP, carotid Doppler, ECG, echo.
- Artery: amaurosis fugax first; CRAO = poor acuity + RAPD; BRAO = field defect. Pale retina, cherry red spot (choroid supplies fovea), pale disc, thin vessels.
- Artery emergency: lower IOP (massage, mannitol, paracentesis); nitrates or 5% CO2.
- Vein: CRVO = poor acuity + RAPD; flame haemorrhages, cotton wool spots, disc swelling, macular oedema, tortuous veins. BRVO: field loss or none.
- Vein treatment: anti-VEGF for macular oedema; PRP for new vessels.

Age-related macular degeneration

- Drusen = lipid in Bruch's membrane. Risk: age, smoking, UV, family history.
- Dry: gradual central loss; drusen then geographic atrophy. Amsler grid, OCT.
- Wet: choroidal new vessels bleed or leak under the retina and scar; distortion, central scotoma.
- All: Amsler grid, stop smoking, sunglasses, vitamins C and E, zinc, beta-carotene. Wet: anti-VEGF, PDT.

Source: VM 043 Retinal vascular diseases

PART 1: LECTURE

Fundus examination skill lab [VM 043]

Topic: Ophthalmology

Preparation and technique

- Start with WIPER: wash hands, greet, introduce, permission (close approach, dark room, bright light), pain check, expose face and neck, patient seated.
- Inspect for glasses/contact lenses, cataract, pupil abnormality, red eye, discharge, lid asymmetry. Check visual acuity and colour vision before the fundus.
- Dim room; large aperture; focusing wheel at zero with index finger on it. Patient removes glasses (not contact lenses) and looks at a distant object.
- Red reflex from arm's length, one pupil at a time. Absent red reflex: cataract, retinoblastoma.
- Right eye = ophthalmoscope in right hand (and vice versa); other hand on forehead, thumb on eyebrow; move in closer.
- Order: disc (colour, contour, cup:disc) then vessels (tortuosity, dilatation) then periphery (look up, down, right, left) then fovea last (look at the light). Thank, explain, document.

Instruments

- Direct: portable, upright image, no dilatation needed. But small field, no stereopsis, spoilt by media opacity.
- Indirect: wide field, better stereopsis. But inverted image, needs dilatation, lower magnification.
- Slit-lamp biomicroscopy: wide field, better stereopsis; inverted image, needs dilatation and skill.

Optic disc

- Normal: vertically oval, pink, cup 30-50% of disc, well-defined contour.
- Papilloedema: passive non-inflammatory swelling from raised ICP, always bilateral. Pink, cup obliterated, blurred margins.
- Optic neuritis: primary inflammation (MS demyelination, TB, vitamin B deficiency). Same disc look: pink, obliterated cup, blurred margins.
- Optic atrophy (e.g. glaucoma): pale disc, increased cup, clear margins.

Vessels and macula

- Tortuous vessels = retinal vein occlusion; haemorrhages = diabetic retinopathy; aneurysms.
- Macula: yellow (xanthophyll), about 6 mm, 2 disc diameters from the disc. Fovea: depressed dark-red centre, 1 disc diameter (about 1.5 mm), cones only.
- Cherry-red spot = retinal artery occlusion; macular oedema = diabetic retinopathy; drusen = age-related macular degeneration.

Source: Fundus exam student notes; Fundus examination skill lab presentation [VM 043]

PART 1: LECTURE

Leprosy [VM 116]

Topic: Dermatology

Organism and spread

- Mycobacterium leprae: chronic granulomatous infection; obligate intracellular, acid-fast rod, not Gram-stainable. Targets macrophages and Schwann cells.
- Not cultured in vitro (mouse foot pad); reservoir is the nine-banded armadillo. Doubling time 12-14 days; incubation 1-20 years (average 5-10).
- Grows best in cool areas: superficial nerves, skin, upper airway mucosa, anterior chamber of the eye, testes.
- Droplet spread needing prolonged close contact with an untreated patient; about 95% naturally immune. Not spread by casual contact, intact skin, pregnancy or sex.

Clinical types (depend on cell-mediated immunity)

- WHO: paucibacillary = 1-5 lesions; multibacillary = more than 5.
- Tuberculoid: 1-2 sharply defined hypopigmented anaesthetic macules, under 10 cm, dry and hairless; face and extensor limbs. Thick, palpable, sometimes tender nerves; early nerve damage. High CMI, smear negative, not infectious.
- Lepromatous: many bilateral symmetrical, poorly defined lesions (macules to nodules), normal sensation. Leonine facies, madarosis, thick ear lobes, nasal collapse. Glove-and-stocking neuropathy (late). Eyes, testes, liver, kidneys, bones affected. Low CMI, many bacilli, infectious.
- Borderline tuberculoid: 5-20 larger lesions, satellites, asymmetrical, less anaesthesia. Mid-borderline: varied lesions, "Swiss cheese". Borderline lepromatous: widespread symmetrical, normal sensation and hair.

Diagnosis

- Skin smear from ear lobes, elbows, knees and lesions. Biopsy: full thickness from the edge of an active lesion (granulomas, giant cells, nerve infiltration). Lepromatous biopsy: many AFB, fat-laden macrophages, few lymphocytes.
- PCR is the most accurate and specific test.
- Lepromin test is not diagnostic; it shows the type: tuberculoid positive (more than 5 mm), lepromatous no reaction.

Treatment and prognosis

- WHO multidrug therapy prevents resistance; patient no longer infectious within 1-2 weeks.
- Paucibacillary: daily dapsone 100 mg + monthly rifampicin 600 mg, 6 months.
- Multibacillary: add clofazimine (50 mg daily, 300 mg monthly); 12-24 months.
- Side effects: haemolysis in G6PD deficiency, allergy, hepatitis, red urine, red skin, ichthyosis.
- Contacts: single-dose rifampicin cuts paucibacillary disease by 50%.
- Skin lesions resolve; nerve damage is permanent. Address stigma, isolation, anxiety and depression.

Source: Leprosy Updated [VM 116]

PART 1: LECTURE

Filariasis [VM 121]

Topic: Dermatology

Organisms and groups

- Nematodes; 4 serious human species: Wuchereria bancrofti, Brugia malayi, Onchocerca volvulus, Loa loa.
- Lymphatic: W. bancrofti, B. malayi, B. timori. Cutaneous: Loa loa, O. volvulus, M. streptocerca. Body cavity: M. perstans, M. ozzardi.
- Lymphatic filariasis (elephantiasis) is spread by mosquitoes (Aedes, Anopheles, Culex, Mansonia). WHO elimination programme uses mass drug administration.

Lymphatic filariasis

- Adults in lymph nodes and vessels (femoral, epitrochlear). Th2 response, raised IgE and IgG4; Wolbachia bacteria add to inflammation.
- Course: asymptomatic microfilaraemia, then acute adenolymphangitis, then chronic irreversible lymphoedema.
- Acute ADL: sudden fever with painful nodes (inguinal, testis, cord), skin peels within a week. Retrograde lymphangitis distinguishes it from bacterial.
- Filarial fever: fever without adenitis. TPE: dry paroxysmal nocturnal cough, wheeze, weight loss, eosinophilia, diffuse CXR infiltrates.
- Lymphoedema: I pitting, reversible; II not reversible, no skin change; III non-pitting, skin thickening; IV warty nodules (elephantiasis).

Diagnosis

- Night blood film 10 pm to 2 am (Loa loa: 10 am to 2 pm).
- Circulating antigen test (W. bancrofti only); antibodies; PCR mainly research.
- Ultrasound "filarial dance" sign; CXR infiltrates in TPE.

Treatment

- DEC is the drug of choice: 6 mg/kg for 12 days; doxycycline 200 mg/day for 6 weeks may be added.
- Lymphoedema: elevation, compression, bed rest, steroids; skin hygiene to prevent infection. Chyluria: low-fat, high-protein diet with MCT.
- Surgery for hydrocele and scrotal elephantiasis.

Onchocerciasis and loiasis

- Onchocerciasis (river blindness): black fly; 99% sub-Saharan Africa. Pruritus, nodules, leopard skin, corneal fibrosis and blindness.
- Diagnosis: skin snips (Africa: gluteal and thigh); slit lamp; Mazzotti test with DEC. Treat with ivermectin 150 mcg/kg every 3 months; moxidectin (age 12+).
- Loiasis: Calabar swellings; worm under the conjunctiva. Daytime blood film.
- Before DEC, count microfilariae (encephalopathy risk). Below 2500/mL: DEC 8-10 mg/kg/day for 21 days. If higher, treat only if symptomatic, after albendazole or apheresis.

Source: Filariasis [VM 121]

PART 1: LECTURE

Poisoning lecture [VM 030]

Topic: Poisoning and toxicology

First steps

- ABC: airway with C-spine control and gag check; O2/ventilation; IV access, bloods, cardiac monitor. If hypotensive with clear lungs: saline or Ringer bolus.
- Empiric: dextrose, oxygen, naloxone (titrated for opioid respiratory depression), thiamine (Wernicke).
- Opioid triad: coma, respiratory depression, pinpoint pupils; look for needle tracks.
- Anticholinergic: delirium, fever, tachycardia, big pupils, dry flushed skin, retention. Sympathomimetic: same but sweating. Cholinergic: secretions, fasciculations.

Antidotes

- Paracetamol: NAC. OP: atropine and pralidoxime. Iron: deferoxamine. Digoxin: Fab. Methanol: ethanol or fomepizole. Benzodiazepine: flumazenil. Opioid: naloxone.
- Methaemoglobin: methylene blue. CO: hyperbaric O2. Cyanide: cyanide kit. TCA, cocaine, salicylate: sodium bicarbonate. Warfarin: PCC, vitamin K. CCB: calcium; beta-blocker and CCB: glucagon.

Decontamination and elimination

- No induced emesis (never salt water). Lavage not routine; may help late for concretions or slow gastric emptying (salicylate, TCA).
- Activated charcoal 50 g, best within 1 h. Not for metals, alcohols, hydrocarbons (aspiration), corrosives, bowel obstruction.
- Whole bowel irrigation (PEG): large iron, slow-release or enteric-coated tablets, packets. Not with ileus, coma or fits unless airway protected.
- Multiple-dose charcoal: theophylline, salicylate, carbamazepine, dapsone, digoxin, TCA.
- Urinary alkalinisation (urine pH 7-8): salicylates, long-acting barbiturates, haemolysis or rhabdomyolysis. Needs normal kidneys; watch K, lung oedema.
- Dialysable drugs: small Vd, low protein binding, small, water-soluble (alcohols, phenobarbital, theophylline, lithium, salicylate).

Key overdoses

- Paracetamol: toxic at 150-250 mg/kg. Often no symptoms at first; liver damage peaks at 72-96 h. Check level on the nomogram and AST/ALT. NAC (glutathione) within 8 h is nearly 100% protective. Risks: alcohol, malnutrition, enzyme inducers.
- TCA: anticholinergic signs, seizures, hypotension, QRS over 100 ms. Give bicarbonate (first line); MDAC.
- Salicylate: tinnitus, vomiting, hyperventilation, fluid loss. Respiratory alkalosis first, then metabolic acidosis. Alkalinise urine (level over 35 mg/dL). Dialyse if level over 100 mg/dL or lung injury, CNS signs, renal failure, refractory acidosis.
- Digoxin: GI symptoms first; bradycardia, AV block, arrhythmias. High K predicts need for Fab. Fab if K over 5, level over 10 ng/mL at 6 h, 10 mg or more taken, or dangerous arrhythmia. Watch for low K after Fab.
- Body packers: if no leak, give lactulose and observe; leak or obstruction may need surgery.

Source: Basics of drug poisoning lecture [VM 030]

PART 1: LECTURE

Poisoning LGW and cases [VM 030, VM 050]

Topic: Poisoning and toxicology

General approach

- High anion gap: MUDPILES (methanol, uraemia, DKA, paraldehyde, iron, lactic acidosis, ethylene glycol, salicylates).
- Coma cocktail: dextrose, naloxone, thiamine. Then GI decontamination, antidotes, elimination, symptom control.

Scorpion and snake

- Scorpion venom causes an autonomic storm (catecholamines plus acetylcholine). Look for high BP, sweating, tremor, arrhythmia, pulmonary oedema, high K, high glucose, acidosis.
- Observe 6 h; admit if systemic signs. Antivenom best within 4 h (up to 24 h): 3-5 ampoules IV/IM, repeat every 30 min. Same dose for children and adults; skin test first.
- Viper (desert): severe swelling, blisters, necrosis. Causes coagulopathy, DIC, haemolysis, myolysis, renal failure and high K. Low platelets and fibrinogen.
- Cobra (Nile Valley): minimal local signs. Fasciculations, then ptosis, diplopia, dysarthria and respiratory paralysis; conscious, sensation intact.
- First aid: rest, immobilise, remove rings. No incision, suction, ice or shocks. Polyvalent antivenom 3-5 vials for any systemic envenomation, never too late. Tetanus cover.

Organophosphates

- OP inhibits AChE, irreversibly after 24-36 h. Carbamates are reversible and rarely fatal.
- Muscarinic DUMBELS; nicotinic MATCH (fasciculations, tachycardia, hypertension); central confusion, convulsions, coma.
- Ix: red-cell (true) and plasma (pseudo) cholinesterase, ABG, ECG, CXR.
- Atropine (muscarinic only): titrate to dry chest and relieved bronchospasm, not heart rate or pupils. Keep atropinised 1-2 days.
- Atropine toxicity: dry hot skin, fever, big fixed pupils, delirium, tachycardia; cool and give fluids and benzodiazepines.
- Oximes reactivate AChE before aging (first 24-48 h); atropine first.
- Intermediate syndrome (2-3 days): neck flexor, proximal and respiratory weakness from muscle necrosis. Delayed neuropathy (2-3 weeks): permanent.

CO and cyanide

- CO binds Hb 200-250x more than O2 and shifts the curve left. Also binds myoglobin and cytochrome oxidase.
- Throbbing headache, coma, MI, pulmonary oedema; cherry-red skin rare. Delayed neuropsychiatric effects after 2-40 days.
- 100% O2: half-life 5 h to 1 h. Hyperbaric O2 (20 min) if confused, COHb over 25%, or fetal distress. Consider cyanide in fire smoke.
- Cyanide (bitter almonds) blocks cytochrome oxidase: red asphyxia, no cyanosis, venous O2 equals arterial, lactic acidosis.
- Cyanide kit: nitrite makes MetHb to bind CN; thiosulfate forms thiocyanate; hydroxocobalamin forms cyanocobalamin.

Source: LGW Poisoning; Poisoning LGW cases; Case 1-3 student notes and handouts [VM 030]

PART 1: LECTURE

Clinical therapeutics: cardiology [VM 110]

Topic: Use of medicine

Antihypertensives

- Dihydropyridines (amlodipine, nifedipine) act on vessels: hypertension, angina incl. vasospastic, Raynaud; AEs oedema, flushing, gingival hyperplasia. Nimodipine = subarachnoid haemorrhage (prevents vasospasm). Verapamil/diltiazem act on heart: AV block, constipation.
- ACE inhibitors: lower angiotensin II, raise bradykinin, lose efferent constriction (lower GFR). Use: hypertension, HF (lower mortality), diabetic proteinuria. AEs: cough, angioedema, hyperkalaemia, teratogen, rise in creatinine; caution in bilateral renal artery stenosis.
- ARBs (losartan): same benefits, no bradykinin rise; used if ACEi cough. Aliskiren = direct renin inhibitor; avoid in pregnancy.
- Hydralazine: arteriolar dilator, safe in pregnancy; reflex tachycardia, drug-induced lupus. Nitroprusside: cyanide toxicity.

Blood thinners

- Heparin: activates antithrombin (IIa, Xa); monitor PTT; reverse with protamine. HIT type 2 at 5-10 days: stop heparin, give argatroban (fondaparinux safe).
- Warfarin: blocks vitamin K epoxide reductase (II, VII, IX, X, C, S); monitor PT/INR; crosses placenta. Early protein C fall = skin necrosis, so bridge with heparin. Reverse: vitamin K; rapid = FFP/PCC.
- Dabigatran (thrombin) reversed by idarucizumab; apixaban/rivaroxaban (Xa) by andexanet alfa.
- Antiplatelets: aspirin (COX, TXA2), clopidogrel/ticagrelor (P2Y12), eptifibatide (GpIIb/IIIa), cilostazol (PDE, claudication).

Diuretics

- Loop (furosemide): ototoxicity, low K and Mg, alkalosis, gout; lose calcium. Thiazides: lower urinary calcium; hyperglycaemia, lipids, uric acid, calcium. Spironolactone: hyperkalaemia, gynaecomastia. Mannitol: raised ICP/IOP; avoid in HF, anuria.

Angina, lipids, heart failure

- Nitrates: veins > arteries, lower preload; headache, reflex tachycardia; contraindicated with PDE-5 inhibitors, RV infarct, HCM. Ranolazine: late Na current, refractory angina.
- Statins: myopathy (worse with fibrates), raised LFTs. Resins: GI upset. PCSK9 inhibitors raise LDL removal.
- Sacubitril/valsartan for HFrEF; never with ACEi (angioedema). SGLT2i: UTI, candida, euglycaemic DKA.
- Digoxin: Na/K ATPase block; toxicity = GI upset, yellow vision, arrhythmias; worsened by low K, high Ca, low Mg, renal failure, verapamil/amiodarone. Antidote: Fab fragments.

Antiarrhythmics

- IA: QT, lupus (procainamide). IC (flecainide): contraindicated post-MI/structural disease. II: beta-blockers. III: amiodarone (check PFT, LFT, TFT). IV: verapamil/diltiazem. Adenosine = SVT drug of choice. Magnesium = torsades, digoxin toxicity. Ivabradine = funny channel; stable angina, HFrEF.

Source: Clinical Therapeutics Cardiology [VM 110] (OCR text)

PART 1: LECTURE

Antiplatelet and anticoagulant therapy [VM 034]

Topic: Use of medicine

Categories and heparins

- Primary prophylaxis = prevent first clot (low-dose heparin after surgery). Therapeutic = treat established DVT to stop propagation/embolism. Secondary prophylaxis = prevent recurrence (warfarin after VTE).
- Heparin accelerates antithrombin against Xa and IIa. Monitor UFH with APTT; LMWH with anti-Xa assay (usually not needed).
- LMWH: longer half-life, once-daily SC, less bleeding, simple dosing, outpatient DVT treatment by weight, HIT very rare.
- UFH preferred: high bleeding risk (renal impairment), need for rapid reversal, massive PE after thrombolysis, LMWH failure. Overdose: stop infusion (half-life 1 h); urgent reversal = protamine.
- Complications: bleeding, HIT, osteoporosis with long use.

Warfarin

- Oral, long half-life; full effect after 3-5 days, so continue heparin until then. Teratogenic. Many drug interactions (antibiotics, fluconazole, aspirin).
- Monitor PT/INR. First DVT/PE: target 2.5 (2-3). Recurrence while therapeutic: target 3.5 (3-4).
- Bleeding risk rises sharply when INR > 5; intracranial bleed most serious.
- INR 6-8, no/minor bleed: stop 24-48 h, recheck, restart lower. INR > 8: stop 48 h, consider vitamin K (oral 2.5 mg or IV 1-2 mg; IV faster, takes at least 6 h).
- Life/sight/limb-threatening bleed: stop warfarin, give PCC (Beriplex) + IV vitamin K. FFP not recommended (incomplete, slow, large volume).

Contraindications and other options

- Absolute: bleeding diathesis, severe hypertension, previous cerebral haemorrhage, CNS trauma, GI bleeding, heparin allergy/HIT. Relative: uraemia, liver disease, alcoholism, SBE. Breastfeeding is not a contraindication.
- IVC filter: anticoagulation contraindicated, recurrence despite adequate anticoagulation, serious complications, free-floating iliofemoral/caval thrombus, very poor lung reserve.
- Thrombolysis: massive PE, phlegmasia cerulea dolens (purple, swollen, painful leg; catheter-directed thrombolysis or thrombectomy).

NOACs and antiplatelets

- Dabigatran (direct thrombin inhibitor); rivaroxaban, apixaban (direct Xa). Advantages: wide therapeutic index, fast onset, predictable, no monitoring. Disadvantages (per lecture): no specific antidote, GI bleeding.
- Fondaparinux: SC pentasaccharide, antithrombin-mediated Xa inhibition only, no thrombin effect; not a NOAC.
- Aspirin 75-81 mg: irreversible COX inhibitor; GI bleeding. Others: ADP receptor and GP IIb/IIIa inhibitors. PDE inhibitors (cilostazol, dipyridamole): raise cAMP, antiplatelet + vasodilator, preferred in PAD.

Source: Antiplatelets and Anticoagulant Therapy [VM 034]; Anticoagulant QA case review

PART 1: LECTURE

Clinical therapeutics: chest [VM 111]

Topic: Use of medicine

Beta-lactams

- Never combine two beta-lactams for the same infection.
- Natural penicillin G/V: Gram-positives (S. pneumoniae, enterococci); penicillin G for neurosyphilis. Anti-staph (nafcillin, oxacillin): MSSA. Amoxicillin/ampicillin: not antipseudomonal. Piperacillin(/tazobactam): Pseudomonas.
- MSSA cover: anti-staph penicillins, BL/BLI combinations, 1st-gen cephalosporin (cefazolin).
- Cephalosporins: 2nd (cefuroxime) Gram-negative, no Pseudomonas; 3rd (ceftriaxone) cross BBB, only ceftazidime covers Pseudomonas; 4th (cefepime) Pseudomonas + MSSA; 5th (ceftaroline) MRSA, not Pseudomonas.
- Carbapenems (imipenem, meropenem, ertapenem): ESBL Gram-negatives, anaerobes, Pseudomonas; poor MRSA. Aztreonam: Gram-negatives incl. Pseudomonas; inhaled in cystic fibrosis.

Other antibiotics and toxicity

- Aminoglycosides (gentamicin, amikacin): Gram-negatives incl. Pseudomonas; streptomycin/amikacin/kanamycin in MDR-TB. Toxic to kidney + ear, neuromuscular block; check audiometry.
- Macrolides (50S): atypicals (Mycoplasma, Chlamydia, Legionella), NTM, H. pylori; QT prolongation.
- Fluoroquinolones (cipro, levo, moxi): QT + tendon. Levofloxacin 750 mg/day antipseudomonal.
- Vancomycin: MRSA, C. difficile, enterococci; no Gram-negatives; kidney + ear, red man syndrome. MRSA: vancomycin, ceftaroline.

Anti-TB (RIPE + S)

- Rifampicin: RNA polymerase; hepatotoxic, enzyme inducer, orange body fluids. Isoniazid: mycolic acid; neuropathy, add B6. Pyrazinamide: hepatotoxic, high uric acid. Ethambutol: optic neuritis, red-green loss. Streptomycin: 30S; ototoxic.

Bronchodilators and anti-inflammatories

- Beta-2 agonists raise cAMP: SABA salbutamol (rescue), LABA formoterol (fast onset), indacaterol/vilanterol 24 h. AEs: tachycardia, tremor, hypokalaemia, tolerance.
- Antimuscarinics (inhaled only): ipratropium (short), tiotropium/umeclidinium (24 h); dry mouth, bitter taste.
- Methylxanthines (oral/IV): PDE block; aminophylline rescue, theophylline maintenance; seizures, arrhythmia, gastritis.
- IV magnesium sulphate 2 g: adjunct in acute severe asthma.
- ICS: oral candidiasis, hoarseness, pneumonia at high dose. Systemic steroids: exacerbations only; infection, osteoporosis, hyperglycaemia, adrenal suppression.
- Roflumilast (oral PDE-4): COPD FEV1 < 50% with exacerbations despite ICS/LABA/LAMA. Azithromycin 250 mg/day long-term: anti-inflammatory. Montelukast: asthma incl. aspirin/exercise-induced.

Inhalers

- MDI (Ventolin) needs coordination; DPI (budesonide) needs inspiratory flow; SMI (Spiriva). 60% use inhalers wrongly. Nebuliser: no coordination, emergencies.

Source: Chest therapeutics [111]; Introduction to Chest Therapeutics [111]

PART 1: LECTURE

Drugs for anaemia [VM 112]

Topic: Use of medicine

Oral iron

- Causes: blood loss (menorrhagia, hookworm, colon cancer), demand (pregnancy, infancy), poor intake or absorption (gastrectomy, coeliac).
- Dose ~200 mg elemental iron daily in 2-3 divided doses for 4-6 months. Absorbed as ferrous form in duodenum; ~25% of ferrous salt absorbed.
- Take 1 h before meals with vitamin C/citrus. Reduced by tea (tannins), phytates, coffee, antacids, milk, calcium, zinc.
- AEs: GI upset, constipation or diarrhoea, metallic taste, teeth staining, dark stools.
- Ferrous sulphate = most common, cheap; ferric salts less bioavailable.

Parenteral iron and toxicity

- Oral is as effective if absorption is normal. Give parenteral if: oral intolerance, haemodialysis on EPO, post-gastrectomy, small bowel resection, proximal IBD, malabsorption, chronic anaemia not maintained orally.
- Iron dextran (IV/deep IM), ferric gluconate, iron sucrose. Anaphylaxis (mainly high-MW dextran): give test dose. Monitor stores; overload treated with deferoxamine.
- Acute poisoning in young children (10 tablets can kill): bloody vomiting/diarrhoea, shock, then metabolic acidosis. Treat: whole bowel irrigation + IV deferoxamine; charcoal does not bind iron.

Folate and vitamin B12

- Folate deficiency: poor diet (alcohol), malabsorption, phenytoin. Also for methotrexate toxicity, haemolytic anaemias.
- Dose 1-5 mg oral daily; pregnancy prophylaxis 400 micrograms daily (neural tube defects). Well absorbed orally; non-toxic.
- B12: hydroxocobalamin preferred (rapid, longer levels). Most deficiency = malabsorption, so IM/deep SC: 100-1000 mcg every 1-2 days for 1-2 weeks, then every 2-3 months for life (every 1-2 weeks for 6 months if neurological signs). Non-toxic.
- Never give folate alone in megaloblastic anaemia: it corrects blood but masks B12 deficiency, letting nerve damage progress.

ESAs and new drugs

- EPO from kidney; low in renal disease, so these patients respond best. Uses: CKD anaemia, AIDS/zidovudine, chemotherapy, before elective surgery or autologous donation, premature infants.
- IV in dialysis, SC otherwise. Target Hb up to but not above 10-12 g/dL; rise no more than 1 g/dL per 2 weeks. Reticulocytes rise ~10 days, Hb 2-6 weeks: no use in acute anaemia. Most need iron.
- Epoetin three times weekly; darbepoetin weekly.
- AEs: hypertension, arthralgia; thrombosis and death if Hb target > 11-12 g/dL.
- Daprodustat: oral HIF-PH inhibitor, raises EPO, lowers hepcidin; dialysis for 4+ months; Hb target not > 11 g/dL.

Source: Drugs for Anaemia with notes [VM 112]

PART 1: LECTURE

Biochemical assay of cholesterol [VM 029]

Topic: Biochemistry

Lipid profile

- Group of tests to assess risk of coronary heart disease, heart attack, hypertension and stroke from atherosclerosis.
- Standard profile: total cholesterol, HDL-C ('good'), LDL-C ('bad'), triglycerides; extended profile adds VLDL-C.
- Sample: serum or plasma after a 12-hour fast (water only).
- After a fatty meal serum is turbid (lipaemic) from chylomicrons; lipoprotein lipase ('clearing factor') clears it within hours.

Lipids and reference values

- Triglycerides: glycerol + 3 fatty acids; main dietary lipid; carried from gut in chylomicrons; excess stored in adipose tissue. Fasting TG < 150 mg/dL.
- Cholesterol: from diet and de novo synthesis; cell membranes, steroid hormones, vitamin D, bile acids. Desirable total < 200 mg/dL.
- Lipoproteins (chylomicrons, VLDL, LDL, HDL) carry hydrophobic lipids in blood.
- LDL carries cholesterol from liver to organs; oxidised LDL deposits in arteries; main drug target. Optimal < 100 mg/dL.
- HDL carries cholesterol from tissues to liver; protective. Low = < 40 mg/dL (men), < 50 mg/dL (women).

Assay principle (enzymatic colorimetry)

- Reagent: cholesterol esterase, cholesterol oxidase, peroxidase, phenol, 4-aminoantipyrine.
- Esterase: cholesterol esters to free cholesterol + fatty acids. Oxidase: cholesterol to H2O2. Peroxidase: H2O2 + phenol + 4-aminoantipyrine to red quinoneimine dye.
- Colour depth is proportional to concentration (more molecules absorb more light). Visible light 380-750 nm; UV < 380; IR > 750.

Practical steps and calculation

- Spectrophotometer = spectrometer (selects wavelength) + photometer (measures intensity). Warm up 15 min.
- Tubes: blank (10 microlitres water), standard (10 microlitres standard), test (10 microlitres serum), each + 1000 microlitres reagent; mix, 5 min at room temperature; read at 546 nm.
- Blank: identical except no analyte; sets zero and removes interference. Wipe cuvette, zero with blank, then read test and standard.
- Concentration (mg/dL) = (A test / A standard) x concentration of standard.
- Checklist: hand gel, gloves, check name/age on sample, check kit expiry. Micropipette: set volume, fit tip, press to first stop, immerse, release; check for bubbles; expel by pressing to second stop.

Source: Biochemical assay of cholesterol [VM 029]

PART 2: TOPIC SUMMARY

Pathology

Blood flow disorders

- Virchow's triad: endothelial injury, stasis or turbulence, hypercoagulability. Stasis = commonest cause of venous thrombosis. Factor V Leiden: young patient with recurrent venous clots.
- Lines of Zahn = thrombus formed in flowing blood during life. Venous thrombi occlude and embolise; arterial thrombi form on ruptured plaques.
- Infarct = coagulative necrosis (brain: liquefactive). Pale wedge: heart, spleen, kidney. Red: lung, dual supply, venous occlusion.
- Nutmeg liver = right heart failure. Dry gangrene: artery only, clear line; moist: artery and vein, no line.

Atherosclerosis, aneurysm, MI

- Fatty streak (earliest, reversible) then fibrous plaque; foam cells = macrophages full of oxidised LDL. Commonest site: lower abdominal aorta.
- AAA: below renal arteries; operate if over 5 cm or enlarging. Berry: circle of Willis, SAH. Charcot-Bouchard: hypertension. Dissection: hypertension, Marfan, not atherosclerosis.

MI age	Finding
1-3 days	Coagulative necrosis, neutrophils
3-7 days	Yellow centre, red rim; rupture risk 2-4 days
10-14 days	Granulation tissue
Over 2 months	Dense scar; late ventricular aneurysm

Heart muscle and hypertension

- Dilated cardiomyopathy (commonest): systolic failure; genetic, alcohol, anthracyclines, past myocarditis.
- Hypertrophic: asymmetric septum, myocyte disarray; sudden death in young people on exercise. Restrictive: amyloid (Congo red, apple-green birefringence). Myocarditis: Coxsackie, T-lymphocytes.
- Benign hypertension: hyaline arteriolosclerosis, small granular kidneys. Malignant: fibrinoid necrosis, onion-skin arterioles, flea-bitten kidneys.

Lung and TB

- Centriacinar emphysema: smokers, upper lobes. Panacinar: alpha-1 antitrypsin, lower lobes. Asthma: Curschmann spirals, Charcot-Leyden crystals. Silica and coal: upper lobes; asbestosis: lower lobes, asbestos bodies. Sarcoid: non-caseating granulomas.
- Lobar pneumonia: congestion, red then grey hepatization, resolution. Abscess: aspirated mouth anaerobes. Aspergilloma: fungus ball in an old cavity.
- TB granuloma: caseation, epithelioid cells, Langhans giant cells. Primary = Ghon complex (subpleural focus + hilar nodes); secondary = apical cavity; miliary foci under 2 mm.

Lung tumours

Type	Clue
Squamous	Central, smoker, keratin pearls, high calcium
Adenocarcinoma	Peripheral, women and non-smokers, TTF-1 positive
Small cell	Central, smoker, SIADH, ACTH; chemotherapy, no surgery
Mesothelioma	Asbestos, calretinin positive, TTF-1 negative

PART 2: TOPIC SUMMARY

Radiology

Chest X-ray basics

- Check technique: AP film and poor inspiration make the heart look big. Adequate film: over 6 anterior or 10 posterior ribs above the diaphragm.
- Silhouette sign: right heart border lost = right middle lobe; left heart border lost = lingula.
- Cardiomegaly: CTR over 0.5. Mediastinum over 8 cm: think dissection.
- Heart failure ABCDE: alveolar (bat wing) oedema, Kerley B lines, cardiomegaly, dilated upper lobe vessels, effusion.
- Hidden areas: apices, behind the heart, under the diaphragm (free air).

Finding	Key sign
Consolidation	Air bronchogram
Collapse	Opacity, volume loss, shift towards; left lower lobe = sail behind heart
Effusion	Blunt costophrenic angle, meniscus
Pneumothorax	Lung edge, no markings beyond; tension = shift away
COPD	Flat diaphragms, wide rib spaces, markings still present

TB imaging

- Primary: consolidation, hilar nodes, effusion, Ghon complex.
- Post-primary: upper zone consolidation, cavity, fibrosis with volume loss; healed = apical fibrosis and calcification.
- Miliary: countless 1-3 mm nodules. Flask-shaped heart = pericardial effusion. Plombage balls = old collapse treatment.

Arterial imaging

- Doppler ultrasound: no radiation. CTA: fast, but X-rays and iodinated contrast. MRA: no radiation. Catheter angiography: invasive.
- Good collaterals with good run-off = chronic occlusion; poor collaterals = acute.
- SFA occlusion: profunda femoris links to genicular branches; treat by angioplasty and stent.
- Fusiform = spindle all round; saccular = one-sided sac.

Venous imaging (DVT)

- Duplex ultrasound = test of choice. Key sign: vein does not compress.
- Acute clot: wide vein, dark thrombus. Chronic: narrow vein, bright thrombus.
- Venography: old gold standard (filling defect). CT venography for iliac veins and IVC; the slide also calls it the choice for the post-phlebitic limb.

Pulmonary embolism

- CTPA first: filling defect in a pulmonary artery; negative CTPA ends the work-up. Pregnancy test first; not needed if Wells unlikely and D-dimer negative.
- V/Q: under 40, contrast allergy, renal failure; normal scan excludes PE; hard to read if CXR abnormal.
- CXR often normal (Hampton's hump). Echo: right heart strain in massive PE. Leg Doppler if high probability but CTPA negative.
- IVC filter in massive DVT to prevent PE.

PART 2: TOPIC SUMMARY

Mental health

Disturbed behaviour

- Rule out organic causes first: delirium, pain, intoxication, withdrawal. Delirium tremens: confusion, hallucinations, 5-10% mortality.
- Warning signs: angry face, restlessness, loud speech, threats, blocking exits.
- Steps: de-escalation, time out (max 15 min), rapid tranquilisation, restraint, seclusion (last resort, review every 2 h).
- Rapid tranquilisation: offer oral first. Lorazepam 1-2 mg, plus haloperidol 5-10 mg if psychotic. Observations every 5-10 min for 1 h; ECG with haloperidol.

Side effect	Action
Acute dystonia	Procyclidine
Akathisia	Reduce drug, propranolol
NMS (fever, rigidity, high CK)	Stop drug, dantrolene, bromocriptine
Benzodiazepine respiratory depression	Oxygen, flumazenil

Depression, anxiety, OCD, stress

- Depression after a cardiac event: sertraline, avoid tricyclics.
- Panic and generalised anxiety: exclude physical causes; CBT, then SSRI; avoid long-term benzodiazepines.
- OCD: intrusive thoughts known to be one's own (unlike psychosis); CBT with exposure and response prevention, SSRI, clomipramine if SSRI fails.
- PTSD: intrusion, avoidance, negative mood, arousal for over 1 month; CBT, EMDR, SSRIs. Acute stress reaction settles within days.

Drugs and alcohol

- Opioid overdose: pinpoint pupils, slow breathing; naloxone is short-acting, so may re-sedate.
- MDMA: hyperthermia, low sodium. Cocaine: MI, stroke, fits.
- Alcohol withdrawal: chlordiazepoxide reducing regimen, thiamine. Wernicke: confusion, ataxia, eye signs; IV thiamine. Korsakoff: loss of new memory.

Self-harm and suicide

- Over 90% have a psychiatric disorder. Self-harm raises suicide risk 50-100 times, highest within 6 months.
- Intent: planning, note, precautions against discovery. Physical severity does not show intent. Risk scales not recommended.
- Asking about suicide does not cause it. Lithium and clozapine reduce suicide.

Egyptian Mental Health Act (2009)

- Capacity: understand, retain, weigh, communicate; presumed; an unwise choice is not incapacity.
- Involuntary admission: psychiatrist approval, prosecutor told within 24 h; non-psychiatrist may hold 48 h; over 1 week needs two psychiatrists. Emergency treatment without consent up to 72 h. ECT only under anaesthesia with written consent.

Other

- Critically ill: ABCDE, treat as found, reassess. Learning disability: IQ under 70 plus adaptive deficit from childhood. Beware diagnostic overshadowing; look for pain. Down's: early dementia. Fragile X: commonest inherited cause in boys.

PART 2: TOPIC SUMMARY

Ophthalmology

Diabetic retinopathy

- Commonest cause of blindness in working-age adults; silent until late, so annual retinal check.

Stage	Signs	Treatment
Non-proliferative	Microaneurysms, dot and blot haemorrhages, hard exudates	Glucose and BP control
Severe NPDR	Cotton wool spots, venous beading, IRMA	PRP if high risk
Proliferative	New vessels (VEGF); vitreous haemorrhage, detachment	PRP, vitrectomy
Macular oedema	Any stage; commonest cause of vision loss	Anti-VEGF

- PRP costs peripheral and night vision.

Hypertensive retinopathy

- Grade 1 slight arterial narrowing; 2 obvious narrowing; 3 plus haemorrhages or exudates; 4 plus disc swelling.

Retinal vessel occlusion (sudden painless loss of vision)

- Artery: pale retina, cherry red spot, thin vessels; often amaurosis fugax first. Emergency: lower eye pressure (massage, mannitol, paracentesis).
- Vein: tortuous veins, flame haemorrhages, cotton wool spots, disc swelling. Anti-VEGF for oedema, PRP for new vessels.
- Work-up: BP, lipids, glucose, ESR/CRP, carotid Doppler, ECG, echo.

Macular degeneration

- Drusen. Dry: slow central loss. Wet: new vessels bleed, distortion; anti-VEGF. All: Amsler grid, stop smoking.

Fundus examination

- Right eye: scope in right hand, your right eye. Red reflex first; absent = cataract or retinoblastoma.
- Order: disc, vessels, periphery, fovea last.
- Direct scope: upright image, no dilatation, small field. Indirect: wide field, inverted image, needs dilatation.

Disc	Look
Normal	Pink, oval, cup 30-50%
Papilloedema	Pink, blurred, no cup; bilateral, raised ICP
Optic neuritis	Same look; inflammatory (MS)
Optic atrophy	Pale, large cup, clear edge

PART 2: TOPIC SUMMARY

Dermatology

Leprosy

- Mycobacterium leprae: acid-fast, cannot be cultured; infects skin and nerves in cool areas. Droplet spread after long close contact; about 95% are immune.
- Type depends on cell-mediated immunity.

	Tuberculoid	Lepromatous
Lesions	1-2, sharp edge, hypopigmented	Many, symmetrical, poorly defined
Sensation	Lost	Normal (late glove-and-stocking)
Nerves	Thick, damaged early	Late
Bacilli	Few, not infectious	Many, infectious
Lepromin test	Positive	Negative

- Lepromatous extras: leonine face, lost eyebrows, thick ear lobes.
- WHO: paucibacillary 1-5 lesions, multibacillary over 5.
- Diagnosis: skin smear, biopsy from active edge; PCR most accurate. Lepromin shows type, not diagnosis.
- Treatment: paucibacillary = dapsone daily + rifampicin monthly for 6 months. Multibacillary = add clofazimine, 12-24 months. Not infectious after 1-2 weeks. Nerve damage is permanent.
- Dapsone: haemolysis in G6PD deficiency. Clofazimine: red skin.

Filariasis

- Lymphatic (W. bancrofti, Brugia): mosquito; fever with painful nodes and retrograde lymphangitis, then chronic lymphoedema up to elephantiasis.
- Tropical pulmonary eosinophilia: night cough, wheeze, eosinophilia.
- Diagnosis: night blood film (10 pm to 2 am); antigen test; ultrasound "filarial dance".
- Treatment: DEC 6 mg/kg for 12 days, with or without doxycycline. Lymphoedema: elevation, compression, skin hygiene. Surgery for hydrocele.
- Onchocerciasis: black fly, itchy nodules, leopard skin, river blindness; skin snips; ivermectin.
- Loa loa: Calabar swellings, eye worm, daytime film. Count microfilariae before DEC (encephalopathy risk).

PART 2: TOPIC SUMMARY

Poisoning and toxicology

First steps

- ABC, then coma cocktail: dextrose, naloxone, thiamine. No induced vomiting.
- Charcoal 50 g, best within 1 h; not for metals, alcohols, hydrocarbons, corrosives.
- Whole bowel irrigation: iron, slow-release tablets, packets. Urine alkalinisation: salicylate. High anion gap: MUDPILES.

Toxidrome	Clues
Opioid	Coma, slow breathing, pinpoint pupils
Anticholinergic	Delirium, fever, big pupils, dry flushed skin, retention
Sympathomimetic	Same, but sweating
Cholinergic	Secretions (DUMBELS), fasciculations

Antidotes

- Paracetamol: NAC. Opioid: naloxone. Benzodiazepine: flumazenil. Organophosphate: atropine + pralidoxime. Iron: deferoxamine. Digoxin: Fab. Methanol: fomepizole or ethanol. TCA and salicylate: bicarbonate. Methaemoglobin: methylene blue. Beta-blocker or CCB: glucagon. Warfarin: PCC, vitamin K.

Key overdoses

- Paracetamol: toxic at 150-250 mg/kg; few early symptoms, liver damage peaks at 72-96 h. Use the nomogram; NAC within 8 h is nearly 100% protective.
- TCA: QRS over 100 ms, fits, low BP: bicarbonate first.
- Salicylate: tinnitus, hyperventilation; respiratory alkalosis then metabolic acidosis. Dialyse if over 100 mg/dL.
- Digoxin: vomiting, bradycardia, heart block; high K predicts need for Fab.

Organophosphates

- Block cholinesterase: muscarinic DUMBELS, nicotinic fasciculations and weakness, confusion, fits.
- Atropine until the chest is dry (not by heart rate or pupils); oxime early, before aging.
- Intermediate syndrome at 2-3 days (neck and breathing weakness); delayed neuropathy at 2-3 weeks.

Bites and stings

- Scorpion: autonomic storm (high BP, sweating, lung oedema); antivenom best within 4 h, same dose for children.
- Viper: swelling, blisters, bleeding (DIC). Cobra: little local sign, ptosis then breathing paralysis.
- No incision, suction or ice. Antivenom is never too late.

CO and cyanide

- CO: headache, coma; 100% O2 cuts half-life from 5 h to 1 h; hyperbaric if confused, COHb over 25% or fetal distress. Late neuropsychiatric effects.
- Cyanide: bitter almonds, no cyanosis, lactic acidosis; hydroxocobalamin, nitrite, thiosulfate.

PART 2: TOPIC SUMMARY

Use of medicine

Cardiovascular drugs

- ACE inhibitors: cough, angioedema, high K, teratogenic; ARB if cough. Never ACEi with sacubitril/valsartan.
- Dihydropyridines: ankle oedema, flushing. Verapamil/diltiazem: heart block, constipation. Hydralazine: safe in pregnancy.
- Nitrates: never with PDE-5 inhibitors. Loop: low K, ototoxic. Thiazide: high glucose, urate, calcium. Spironolactone: high K, gynaecomastia.
- Digoxin toxicity: GI upset, yellow vision, arrhythmia; worse with low K; antidote Fab.
- Amiodarone: check lung, liver, thyroid. Adenosine for SVT; magnesium for torsades. Flecainide: not after MI.

Anticoagulants

Drug	Target	Monitor	Reversal
Unfractionated heparin	Antithrombin (IIa, Xa)	APTT	Protamine
Warfarin	Vitamin K factors	INR 2-3	Vitamin K; PCC if serious bleed
Dabigatran	Thrombin	None	Idarucizumab
Apixaban, rivaroxaban	Xa	None	Andexanet alfa

- One slide says NOACs have no specific antidote; the cardiology slides list the two above.
- LMWH: once daily, less bleeding, HIT rare. UFH preferred in renal failure or when fast reversal is needed.
- Warfarin takes 3-5 days: overlap with heparin. Target INR 2.5; 3.5 if recurrence on treatment. INR over 8: stop, consider vitamin K. FFP not recommended.
- HIT (5-10 days): stop heparin, give argatroban. Antiplatelets: aspirin (COX), clopidogrel (P2Y12), cilostazol (claudication).

Chest drugs

- Never combine two beta-lactams. Pseudomonas: piperacillin/tazobactam, ceftazidime, cefepime, carbapenems, aminoglycosides. MRSA: vancomycin, ceftaroline.
- Aminoglycosides, vancomycin: kidney and ear. Macrolides, quinolones: long QT; quinolones: tendons.
- TB: rifampicin orange fluids, enzyme inducer; isoniazid neuropathy (give B6); pyrazinamide high urate; ethambutol red-green vision.
- SABA rescue; LABA, LAMA maintenance; IV magnesium 2 g in severe asthma; ICS cause oral thrush. 60% use inhalers wrongly.

Anaemia

- Oral iron: ferrous sulphate, about 200 mg elemental daily for 4-6 months, before food with vitamin C; tea and antacids reduce it.
- Child iron overdose: deferoxamine, bowel irrigation; charcoal useless.
- B12: IM hydroxocobalamin for life. Never folate alone in megaloblastic anaemia. Pregnancy folate 400 micrograms.
- IV iron if oral fails or malabsorption; test dose (dextran anaphylaxis).
- EPO in CKD: Hb target not above 10-12 g/dL; needs iron; no use in acute anaemia.

PART 2: TOPIC SUMMARY

Biochemistry

Lipid profile

- Assesses risk of coronary disease and stroke from atherosclerosis: total cholesterol, HDL, LDL, triglycerides (extended profile adds VLDL).
- Sample: serum or plasma after a 12-hour fast (water only). After a fatty meal serum is milky from chylomicrons; lipoprotein lipase clears it.

Lipid	Target
Total cholesterol	Under 200 mg/dL
LDL ("bad", main drug target)	Under 100 mg/dL
HDL ("good", protective)	Low if under 40 (men), under 50 (women)
Fasting triglycerides	Under 150 mg/dL

- LDL carries cholesterol from liver to tissues; oxidised LDL deposits in arteries. HDL carries it back to the liver.

Assay principle (enzymatic colour test)

- Cholesterol esterase frees cholesterol from esters; cholesterol oxidase makes H2O2; peroxidase joins H2O2 with phenol and 4-aminoantipyrine to form a red dye.
- Colour depth is proportional to concentration. Read at 546 nm (visible light 380-750 nm).

Practical steps

- Spectrophotometer: warm up 15 min. Three tubes, each 1000 microlitres reagent plus 10 microlitres of water (blank), standard or serum. Mix, 5 min at room temperature.
- Blank sets zero and removes interference. Wipe cuvette, zero with blank, read standard and test.
- Concentration = (A test / A standard) x standard concentration.
- Micropipette: press to first stop to draw, second stop to expel; check for bubbles. Check patient name and kit expiry.